



ISO 11783-11
Mobile Data Element Dictionary
DDE Request Form



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DDE Supplement / Attachment

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Tramline Control

1 General

1.1 General Overview

This specification defines basic methods and requirements for a Tramline Control system consisting of an implement (e.g. seeder, planter) and a Task Controller.

1.2 Terms and definitions

1.2.1 Bout

Bouts are the parallel Swathes with a Working Width going from one side to the other side of the field, as in mowing or seeding.

1.2.2 Tramline

Tramlines are a path of parallel tire lines on the field where no crops are growing, that allow the farmers to drive through the field for follow up processes (e.g. fertilizing or spraying) without causing any damage on surrounding crops.

1.2.3 Symmetric Tramlines

Tramline tracks are located in one Bout.

1.2.4 Asymmetric Tramlines

Tramline tracks are divided over two Bouts.

1.2.5 Tramline Rhythm

Total number of seeding Bouts which are required to repeat the Tramline pattern. This is always a positive number > 0 .

1.2.6 Bout Track Number

Repeating Track Number of a Bout, starting from 1 to the Tramline Rhythm Number.

1.2.7 Guidance Track Number

Number of the Track provided by the Task Controller. This Track Number is a unique number. The track number 0 is defined by the user. The user can define any existing guidance line as track 0. The Tracks on the right-hand side of track 0, in direction A to B, will have ascending positive numbering.
The Tracks on the left-hand side of track 0, in direction A to B, will have descending negative numbers. This Track Number 0 is the starting point of the work for the rest of the field.

Tramline Control

1.3 Tramline Control Levels

With the introduction of GNSS in the agricultural industry, Implement devices may use the GNSS information as input for the tramlining track calculations instead of the signal from the track markers.

This document provides the specifications for three different levels in standardization for tramlining:

- Level 1 Task Controller, Implement calculates tramlines (minimum setup)
- Level 2 Task Controller, Implement calculates tramlines (extended setup)
- Level 3 Task Controller calculates tramlines

All related parameters are defined as DDIs in chapter 2.22.

1.3.1 Level 1 – Task Controller, Implement calculates tramlines (minimum setup)

Level 1 describes the minimum required configuration for Tramline Control:

The implement calculates the tramline tracks.

There is only a minimum communication between the Task Controller and the implement.

The following table shows the list of DDIs to be used for a Level 1 Tramline Control.

In case an optional DDI is provided in the DDOP of the implement, the TC is required to send value commands with appropriate values to the implement. The Actual Trimline Condensed Work State DDIs may be sent by the implement, the TC may visualize the tramlines.

DDI (dec)	Description	Part of DDOP
505	Tramline Control Level	required
506	Setpoint Tramline Control Level	required
515	Tramline Control State	required
507	Tramline Sequence Number	required
508	Unique A-B Guidance Reference Line ID	required
509	Actual Track Number	required
510	Track Number to the right	required
511	Track Number to the left	required
512	Guidance Line Swath Width	optional
513	Guidance Line Deviation	optional

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514	GNSS Quality	optional
518, 603-617	Actual Tramline Condensed Work State 1-16,17-256	optional

Table 1: Level 1 DDIs

1.3.2 Level 2 – Task Controller, Implement calculates tramlines (extended setup)

Level 2 extends Tramline Control functionality:

The implement is responsible for the calculation of the tramline tracks.

Level 2 extends the functionality of Level 1 with the following functions:

Support of asymmetric tramline rhythm in all field conditions. With tramline configuration information from the Implement, the Task Controller is able to create the O_R and O_L guidance lines automatically in case an asymmetric rhythm is transferred to a symmetric rhythm.

Restore settings: the implement may send the tramline configuration information to the Task Controller, so the information can be stored with guidance line resp. guidance pattern and reused later.

The following table shows the list of DDIs to be used for a Level 2 Tramline System.

In case an optional DDI is provided in the DDOP of the implement, the TC is required to send value commands with appropriate values to the implement.

DDI (dec)	Description	Part of DDOP
505	Tramline Control Level	required
506	Setpoint Tramline Control Level	required
515	Tramline Control State	required
507	Tramline Sequence Number	required
508	Unique A-B Guidance Reference Line ID	required
509	Actual Track Number	required
510	Track Number to the right	required
511	Track Number to the left	required
512	Guidance Line Swath Width	optional

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513	Guidance Line Deviation	optional
514	GNSS Quality	optional
70	Maximum Working Width	optional
516	Tramline Overdosing Rate	optional
518, 603-617	Actual Tramline Condensed Work State 1-16,17-256	required
666	Working Direction	optional
667	Distance between Guidance Track 0 _R and 1	required
668	Distance between Guidance Track 0 _R and 0 _L	required
669	Track Number Shift	optional
670	Tramline Primary Working Width	optional
671	Tramline Primary Tire Width	optional
672	Tramline Primary Wheel Distance	optional
673	Tramline Secondary Working Width	optional
674	Tramline Secondary Tire Width	optional
675	Tramline Secondary Wheel Distance	optional

Table 2: Level 2 DDIs

1.3.3 Level 3 – Task Controller calculates tramlines

The Task Controller is responsible for calculating the position of the tramline tracks in the field. Level 3 requires full Tramline Control logic as part of the Task Controller. The implement receives only setpoints.

The following table shows the list of DDIs to be used for a Level 3 Tramline Control.

In case an optional DDI is provided in the DDOP of the implement, the TC is required to send value commands with appropriate values to the implement.

DDI (dec)	Description	Part of DDOP
505	Tramline Control Level	required

Tramline Control

506	Setpoint Tramline Control Level	required
515	Tramline Control State	required
70	Maximum Working Width	required
516	Tramline Overdosing Rate	optional
517, 618-632	Setpoint Tramline Condensed Work State 1-16, 17-256	required
518, 603-617	Actual Tramline Condensed Work State 1-16,17-256	required

Table 3: Level 3 DDIs

2 Tramline Control system requirements

2.1 Task Controller as Tramline Controller

If Task Controller and Implement are both TC version 4 or later, the SetValueAndAcknowledgeCommand (0x0A) shall be used instead of the normal ValueCommand (0x03) to send values on the network. Previous versions shall use the ValueCommand (0x03).

2.1.1 Storage of implement tramline parameters in the Task Controller

For tramline control level 2, the Task Controller shall store the tramline parameters (see Table 4:) depending on the active guidance line or guidance pattern.

The task controller shall send the stored values depending on the active Guideline/Guidance Pattern with the first tramline sequence to the implement, see figure 1. The implement may use these values for the tramline calculation.

To save new values or changed values, the Task Controller shall request the values from the implement. This request shall be sent after the completion of the first Tramline sequence and before sending the second Tramline sequence, see Figure 2.

Tramline Control

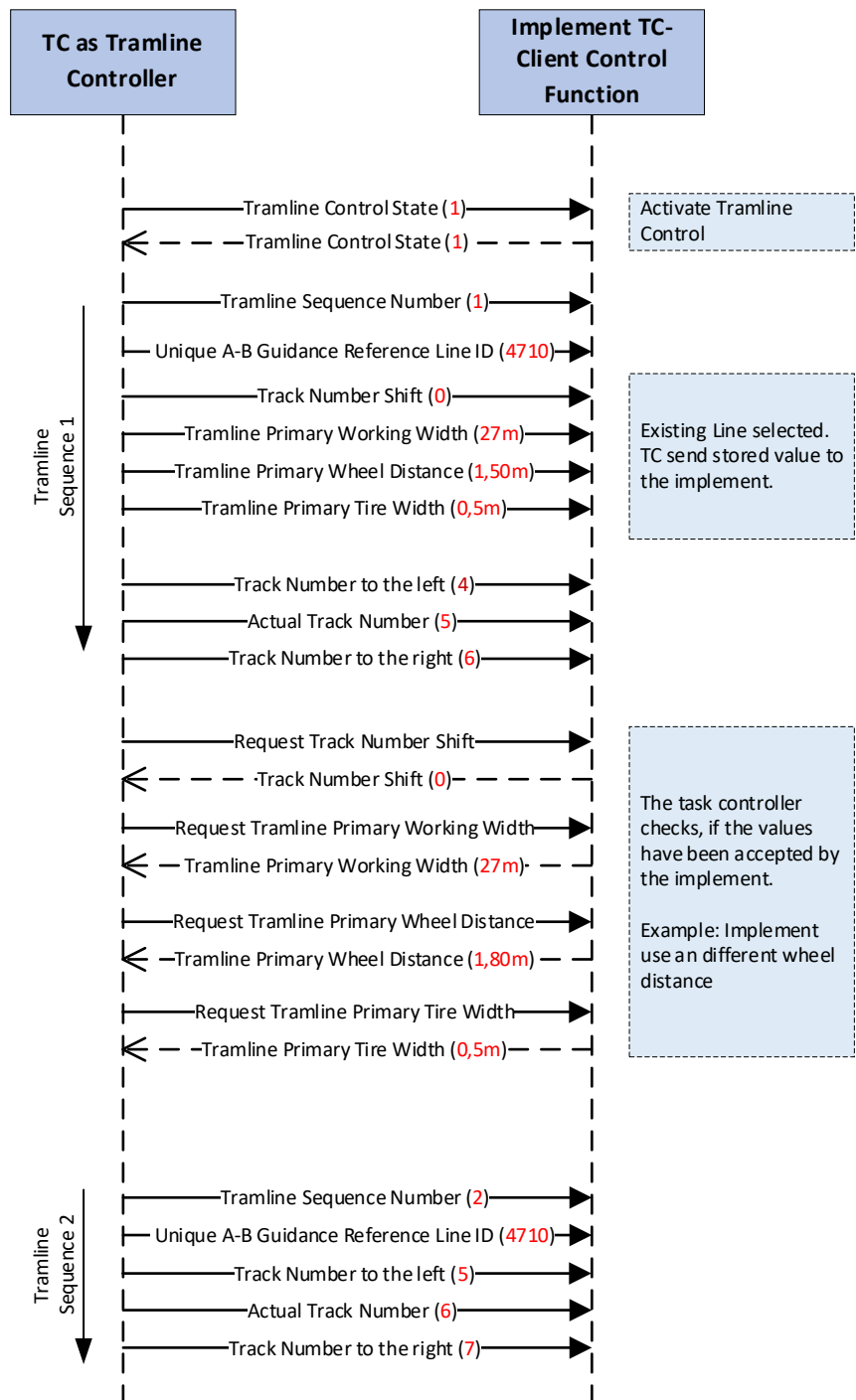


Figure 1: Sequence diagram: Restore of saved values.

Tramline Control

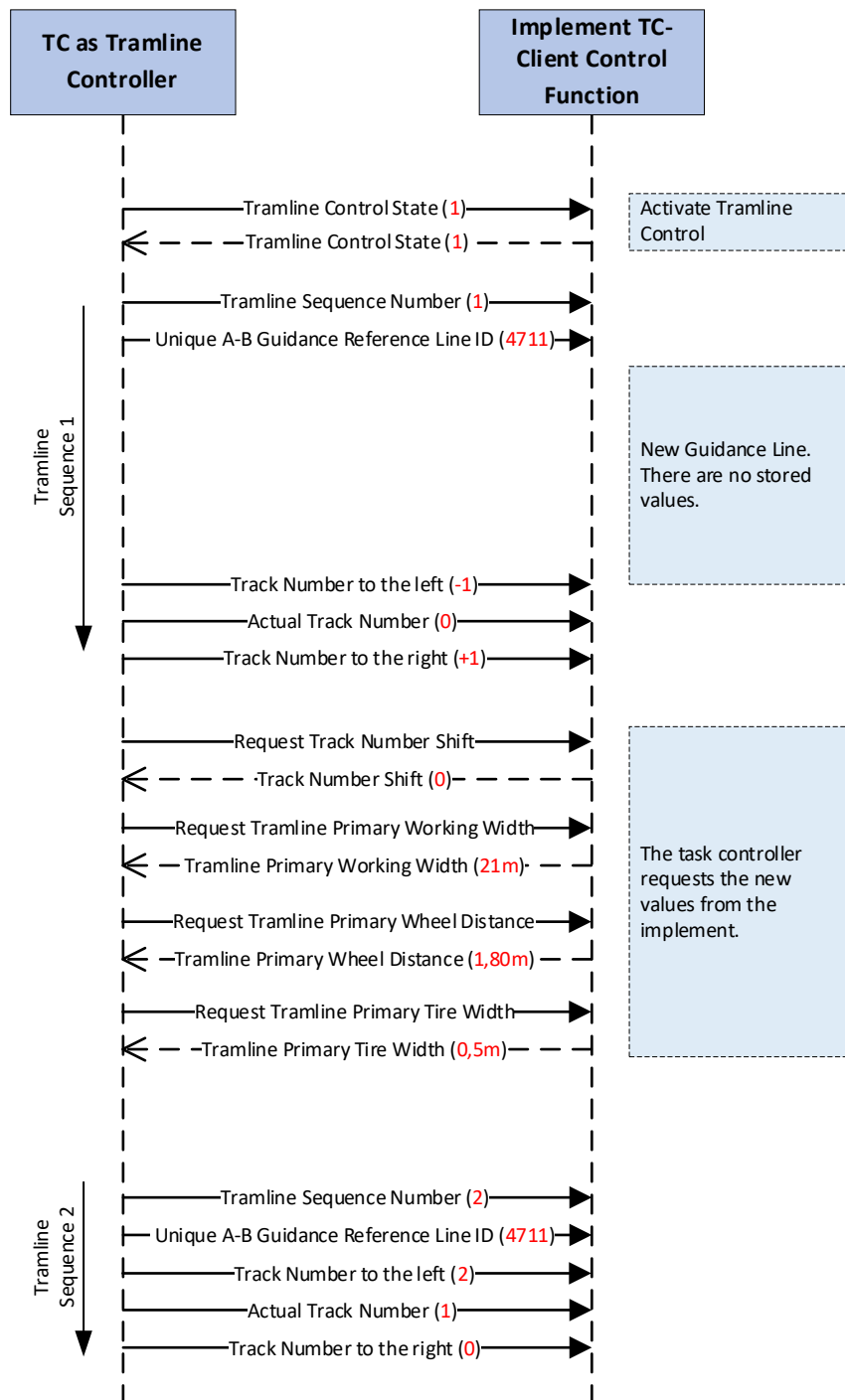


Figure 2: Sequence diagram: Requests from new values

Tramline Control

DDIs listed in table 4 may be part of the DDOP.

DDI (dec)	Description
516	Tramline Overdosing Rate
666	Working Direction
669	Track Number Shift
670	Tramline Primary Working Width
671	Tramline Primary Tire Width
672	Tramline Primary Wheel Distance
673	Tramline Secondary Working Width
674	Tramline Secondary Tire Width
675	Tramline Secondary Wheel Distance

Table 4: Possible tramline parameters to be stored by the TC.

2.2 Description of the DDIs

This chapter describes the DDIs

2.2.1 DDI 515 Tramline Control State

The Tramline Control State has the same purpose and definition like the Section Control State DDI 160.

The value definitions are:

Byte 1 Bits 0-1 = 00 manual/off
Byte 1 Bits 0-1 = 01 automatic/on
Byte 1 Bits 0-1 = 10 error
Byte 1 Bits 0-1 = 11 undefined/not supported
Byte 1 Bits 2-7 reserved, set to 0.

The DDI shall support the On Change trigger so that the Task Controller is able to get informed when the value gets changed by the Working Set Master. The Task Controller shall activate this trigger when using the DDI. For Tramline Control Level 1 and Tramline Control Level 2, this DDI shall be listed in the DDOP only once.

For Tramline Control Level 3, this DDI shall be listed in the DDOP in every boom with settable tramline sections.

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2.2.2 DDI 505 Tramline Control Level

The Implement shall inform which Tramline Control Levels are supported. The Tramline Control Levels are independent of each other. It is allowed to support for example only Level 3 Tramlining. The value of this DDI shall not change during runtime. Therefore this DDI should be listed in the DDOP as DPT. This DDI shall be listed in the DDOP only once.

Bit 0 = 1 Support Tramline Level 1
Bit 1 = 1 Support Tramline Level 2
Bit 2 = 1 Support Tramline Level 3
Bit 3-7 = 0 Reserved

2.2.3 DDI 506 Setpoint Tramline Control Level

The Task Controller shall send this value to inform the Implement which Tramline Control Level shall be used to operate. In case there is no match between the supported Tramline Control Level on the Task Controller Server side and the Implement, the Task Controller shall inform the Implement by setting the Setpoint Tramline Control Level to 0. This DDI shall be listed in the DDOP only once. This DDI shall be placed in the same device element as DDI 505.

0 No common Level
1 Tramline Control Level 1
2 Tramline Control Level 2
3 Tramline Control Level 3
4-255 Reserved for future Assignment

2.2.4 DDI 70 Maximum Working Width

This is the Working Width of the implement. This value is provided by the implement as its own Maximum Working Width. This value depends on the current setup of the Implement. The implement calculates the resulting tramlines in level 1 and level 2 based on this value.

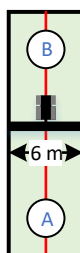


Figure 3: Implement Maximum Working Width

2.2.5 DDI 670 Tramline Primary Working Width

Working Width of the primary tramline vehicle, e.g. crop protection/fertilization vehicle. This is also the distance between the single Tramline Tracks. This value is a manual user input and is provided by the Implement to the Task Controller. If provided in the DDOP, this DDI shall be listed only once.

This value shall be stored by the Task Controller. See also 2.1.1

Tramline Control

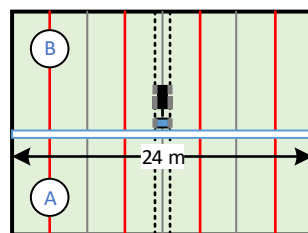


Figure 4: Tramline Working Width

2.2.6 DDI 671 Tramline Primary Tire Width

This is the Width of the largest tire which should fit on the Tramline Track. The distance between the rows besides the Tramline Track shall be wider than the Tire Width to avoid crop damage. If provided in the DDOP, this DDI shall be listed only once.

This value shall be stored by the Task Controller. See also 2.1.1

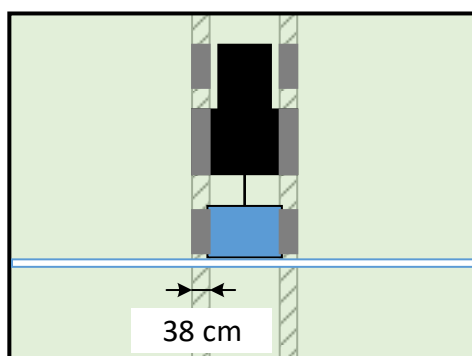


Figure 5: Tramline Tire Width

2.2.7 DDI 672 Tramline Primary Wheel Distance

This is the distance between the centres of the Wheels of the Sprayer/Tractor. If provided in the DDOP, this DDI shall be listed only once.

This value shall be stored by the Task Controller. See also 2.1.1

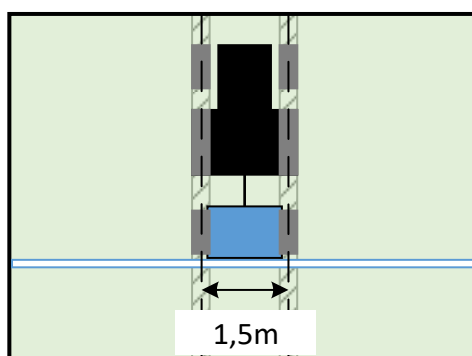


Figure 6: Tramlining Wheel Distance

Tramline Control

2.2.8 DDI 673 Tramline Secondary Working Width

Besides a primary tramlining rhythm e.g., for a crop protection/fertilizing, a second tramlining rhythm may exist. E.g., irrigation or slurry tanker.

The Working Width of these implements are also the distance between the single Tramline Tracks. This value is a manual user input and is provided by the Implement to the Tramline Controller. If provided in the DDOP, this DDI shall be listed only once.

This value shall be stored by the Task Controller. See also 2.1.1

The following picture shows an example with a 6m Seed drill Width, 24m Sprayer Width and a 66m Irrigation Width.

Primary Bout Track No.	1	2	3	4	1	2	3	4	1	2	3	4	1
Secondary Bout Track No.	1	2	3	4	5	6	7	8	9	10	11	1	2
	0	1	2	3	4	5	6	7	8	9	10	11	12

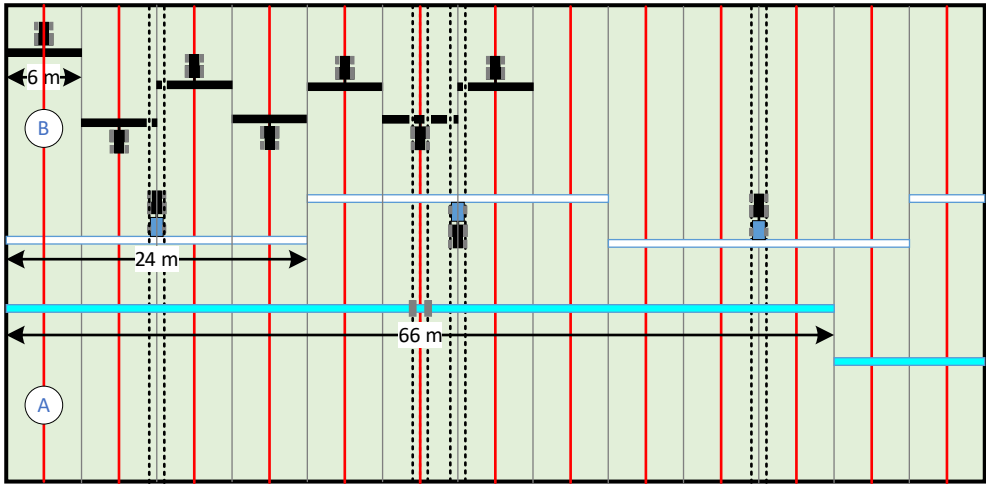


Figure 7: Different Tramline Widths

2.2.9 DDI 674 Tramline Secondary Tire Width

This is the Width of the largest Tire which should fit on the secondary Tramline Track. The distance between the rows besides the Tramline Track shall be wider than the Tire Width to avoid crop damage. If provided in the DDOP, this DDI shall be listed only once.

This value shall be stored by the Task Controller. See also 2.1.1

2.2.10DDI 675 Tramline Secondary Wheel Distance

This is the Distance between the centre of the Wheels of the secondary implement. If provided in the DDOP, this DDI shall be listed only once.

This value shall be stored by the Task Controller. See also 2.1.1

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2.2.11 DDI 666 Working Direction

The Working Direction defines the intended Working Direction in the field and also defines the numbering of the Bouts. If the Working Direction is 1 (working from left to right, compared to AB-reference line) the numbering of the Bouts is also from left to right in ascending order. If the Working Direction is 2 (working from right to left, compared to AB-reference line) the numbering of the Bouts is from right to left in ascending order.

For Working Direction = 1 the numbering of Bouts and Guidance Tracks are increasing from left to right.

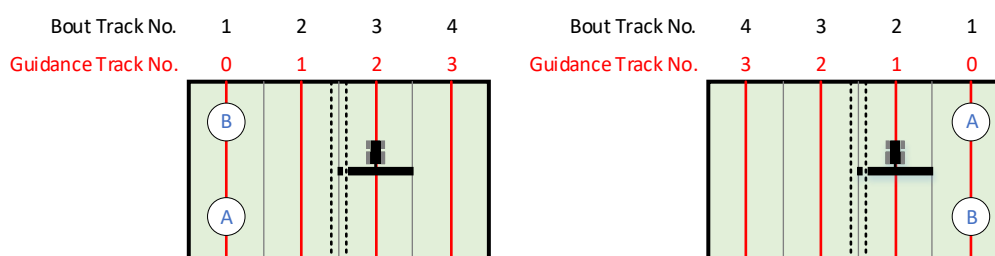


Figure 8: Working Direction = 1 from left to right in A-B direction

For Working Direction = 2 the numbering of Bout Tracks are increasing and the Guidance Tracks are decreasing from right to left.

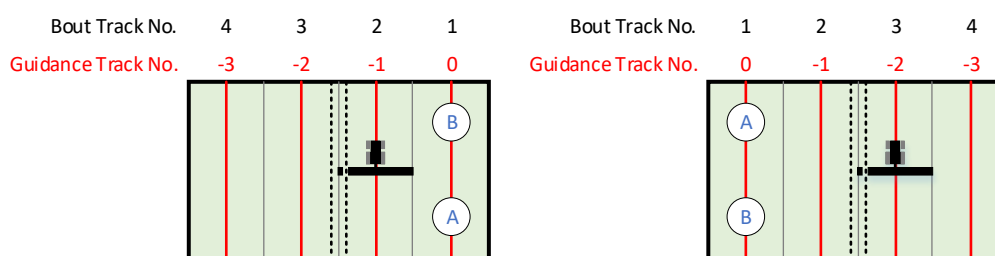


Figure 9: Working Direction = 2 from right to left in A-B direction

Working Direction = 0 means that the direction is unknown. In this case it is up to the implement how to deal with it. The user may be forced to enter a proper value. If provided in the DDOP, this DDI shall be listed only once.

This value shall be stored by the Task Controller. See also 2.1.1.

2.2.12 DDI 508 Unique A-B Guidance Reference Line ID

A field could have more than one Guidance Reference Line. For example, the field could have a Guidance Reference Line for the headland and also another Guidance Reference Line for the main field. To distinguish between the several Guidance Reference Lines, each has an own Guidance Reference Line ID which is stored to the field. This unique ID identifies which Guidance Reference Line is the base for the Tramline calculation. This DDI shall be listed in the DDOP only once.

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2.2.13DDI 509 Actual Track Number

The Actual Track Number is the unique Number of the Guidance Track the Implement is currently located on. This number is provided by the Task Controller. This DDI shall be listed in the DDOP only once.

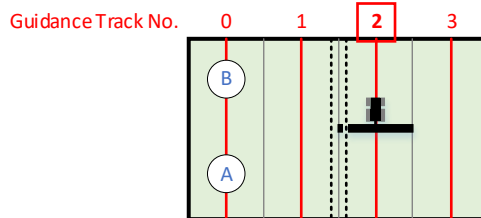


Figure 10: Actual Track Number

2.2.14DDI 510 Track Number to the right

This is the Guidance Track Number to right hand side in direction of Implement orientation. If the Track Number to the right is higher than the Actual Track Number, then the Implement is in the same direction as the A-B reference line. The Implement orientation is independent of the driving direction of the Implement. This value is provided by the Task Controller. This DDI shall be listed in the DDOP only once.

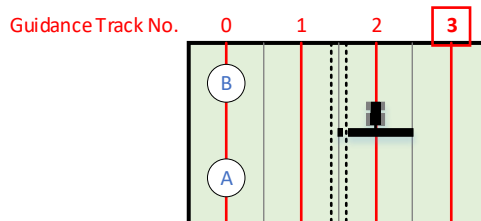


Figure 11: Track Number to the right

2.2.15DDI 511 Track Number to the left

This is the Guidance Track Number to left hand side in Implement orientation. This value may be used in addition to the Track Number to the right for differentiating between the two Guidance Track 0 Numbers (0_R and 0_L). This value is needed, because there are two cases where the Actual Track Number and the Track Number to the right are both 0. This value is provided by the Task Controller. This DDI shall be listed in the DDOP only once.

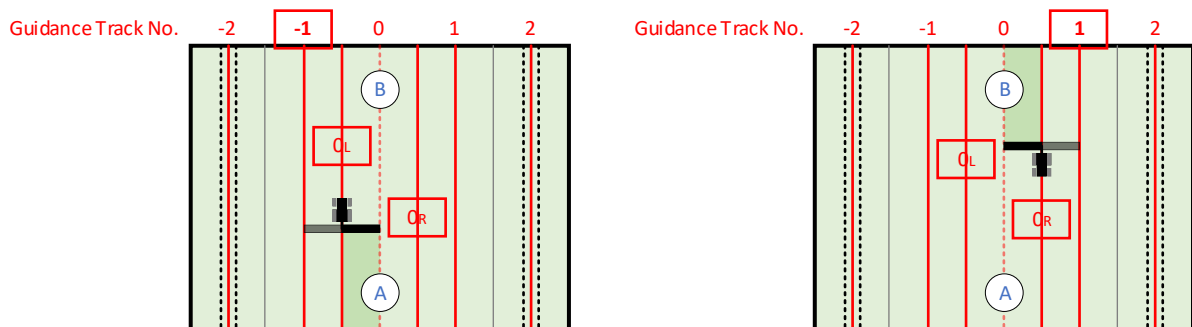


Figure 12: Track Number to the left

Tramline Control

The following table contains some possible combinations for the Track Numbers.

Use case implement is on Track and orientation:	0 _{LAB}	0 _{LBA}	0 _{AB}	0 _{BA}	0 _{RAB}	0 _{RBA}
Actual Track Number	0	0	0	0	0	0
Track Number to the right	0	-1	1	-1	1	0
Track Number to the left	-1	0	-1	1	0	1

2.2.16DDI 512 Guidance Line Swath Width

The Swath Width is the distance between two adjacent Guidance Lines in an active Guidance Pattern. The User may prefer to choose a distance between two adjacent Guidance Lines which is a little smaller or bigger than the implement Working Width on Task Controller side. If this setting is changed, the implement may use this information to apply correction on its Tramline calculation. It is up to the task controller manufacturer whether to respond to Guidance Line Swath Width change requests from the device side. If provided in the DDOP, this DDI shall be listed only once.

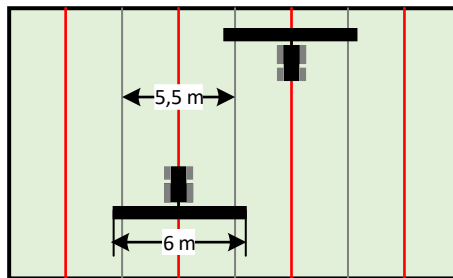


Figure 13: Guidance Swath Width is smaller than Implement Working Width

2.2.17DDI 513 Guidance Line Deviation

This DDI shall be sent from the Task Controller. Looking from the Implement to the Actual Track Number (unique Guidance Track Number), it specifies the Deviation in mm from this Guidance Line and the Device Reference Point which is guided along the Guidance Line. The Guidance Line Deviation is positive when the Guidance Line is located on the right-hand side of the Device Reference Point. The Guidance Line Deviation is negative when the Guidance Line is located on the left-hand side of Device Reference Point. If provided in the DDOP, this DDI shall be listed only once. The TC shall sent the value on change.

The Implement may use this information to inform the operator that a proper Tramline calculation isn't possible from a certain value.

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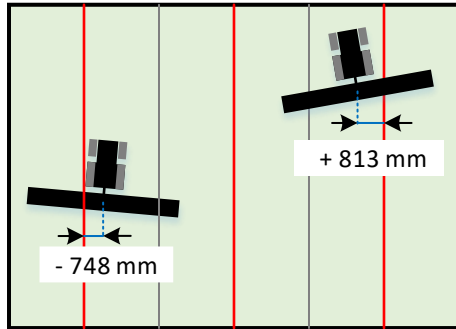


Figure 14: Guidance Line Deviation

2.2.18DDI 514 GNSS Quality

This DDI shall be sent from the Task Controller to the implement. It specifies the quality of the GNSS which was used by the Task Controller. The GNSS receiver may not have a direct communication to the Implement in case of NMEA 0183 receiver. If provided in the DDOP, this DDI shall be listed only once. The TC shall send the value on change.

Definition references NMEA2000 Method GNSS parameter as also mentioned in ISO11783-10.

2.2.19DDI 516 Tramline Overdosing Rate

This DDI specifies the Overdosing Rate for the rows adjacent to the Tramline Tracks. This value is specified in ppm. The value 1.000.000 (100%) is the normal rate. A value > 100% means that a overdosing is applied.

In case of a seeding distance, the implement should calculate the overdosing based on seeds per area and recalculate that value to an overdosing seeding rate.

Background: If the seeding distance is just multiplied with the overdosing rate, the seeding distance will increase which leads to less seed on the field!

Example: In the case the seeding distance is specified the overdosing rate will result in a shorter seed distance. A seeding distance of 20,00 cm with an overdosing rate of 110% will result in a seed distance of 18,18cm for the rows adjacent to the Tramline Tracks.

This value shall be stored by the Task Controller. See also 2.1.1

2.2.20DDI 518, 603-617 Actual Tramline Condensed Work State 1-16, 17-256

Combination of the Actual States of individual Tramline Valves number 1 to 16 into a single Actual Tramline State of their parent DeviceElement. The Actual Tramline Condensed Work State contains the child element Actual Tramline Work States, in the driving direction from left to right, where the leftmost child element Actual Tramline Work State are the 2 lowest significant bits of the Process Data Value.

Each child device elements Actual Tramline Work State is represented by 2 bits and defined as: 00 = do not create a tramline track, 01 = create a tramline track, 10 = error indicator, 11 = not installed/not available.

In total 16 child device element Actual Tramline Work States can be contained in one Actual Tramline Condensed Work State of their parent DeviceElement.

If less than 16 child device element Actual Tramline Work States are available, then the unused bits shall be set to value 11 (not installed).

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In case the implement switches of seeding elements to create the tramline track, the actual section control work state of that element should be set to "on". This avoids that the section control switches on the seeding elements if the tramline track area is re-entered.

2.2.21 DDI 517, 618-632 Setpoint Tramline Condensed Work State 1-16, 17-256

The Setpoint Tramline Condensed Work State DDIs are the control command counterparts to the Actual Tramline Condensed Work States DDIs. The value is a combination of the Setpoint Tramline Valve Work States of individual Tramline Valves number 1 to 16 into a single Setpoint Tramline Work State of their parent DeviceElement. The Setpoint Tramline Condensed Work State contains the child element Setpoint Tramline Work States, in the driving direction from left to right, where the leftmost child element Setpoint Tramline Work State are the 2 lowest significant bits of the Process Data Value.

Each child device elements Setpoint Work State is represented by 2 bits and defined as: 00 = do not create a tramline track, 01 = create a tramline track, 10 = error indicator, 11 = no change. In total 16 child device element Setpoint Tramline Work States can be contained in one Setpoint Tramline Condensed Work State of their parent DeviceElement.

If less than 16 child device element Setpoint Tramline Work States are available, then the unused bits shall be set to value 11 (no change).

2.2.22 DDI 507 Tramline Sequence Number

In order to ensure that the parameters "Unique A-B Guidance Reference Line ID", "Actual Track Number", "Track Number to the right" and "Track Number the left" are belonging together, the Tramline Sequence Number is needed. This parameter has to be sent from the Tramline Controller to indicate a new Tramline Sequence to the Implement. This number shall start with value 1 and increase on every new Tramline Sequence which is going to be sent. If the Tramline Sequence Number exceeds the range, it shall start from 1 again.

These Parameters are only allowed to be sent in a group. If one of the parameters is missing, the Tramline Sequence would be invalid. It is up to the Implement how to behave in case of an invalid Tramline Sequence.

The Unique A-B Guidance Reference Line ID shall be sent as first value after the Tramline Sequence Number and before other values.

It is recommended to send all values belonging to one Tramline Sequence within 500 ms. This DDI shall be listed in the DDOP only once.

Tramline Control

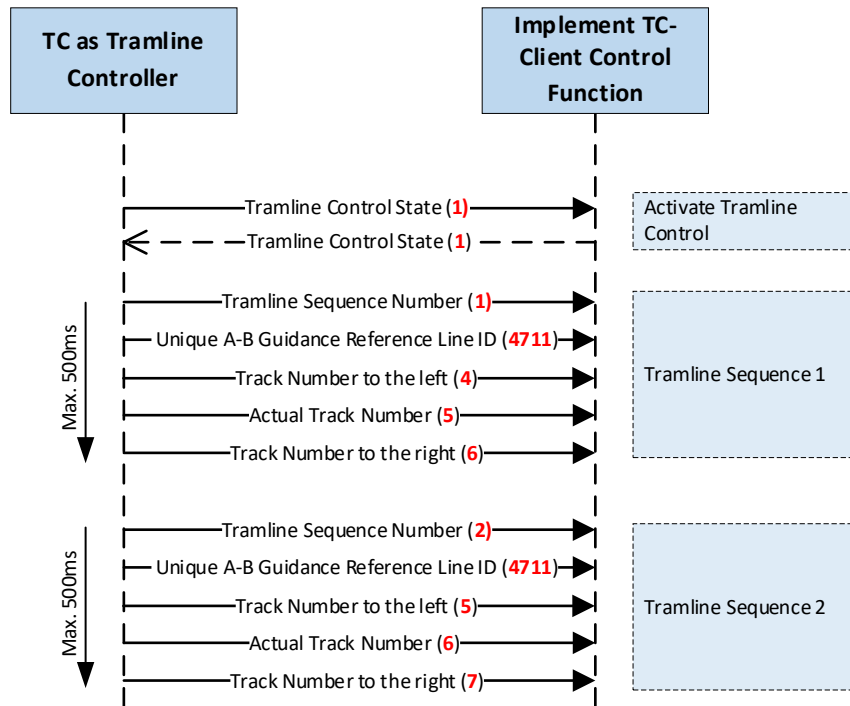


Figure 15: Sequence diagram Tramline Sequence

2.2.23 DDIs 667 + 668 Distances between Guidance Track Number 0_R and 1, and Track 0_R and 0_L

If the user wants to start with a different Working Width than 100%, the Track Number 0 has to be divided in two separate Tracks 0_R and 0_L . Therefore, the Tracks 0_R and 0_L are deviated from the original Track 0. To calculate the correct positions of these Tracks, the distance $d_{0R/1}$ between Track 0_R and 1 and the distance $d_{0R/0L}$ between 0_R and 0_L have to be defined.

This section defines two separate DDIs, one for $d_{0R/1}$ and one for $d_{0R/0L}$ although they are specified together in the same text and pictures.

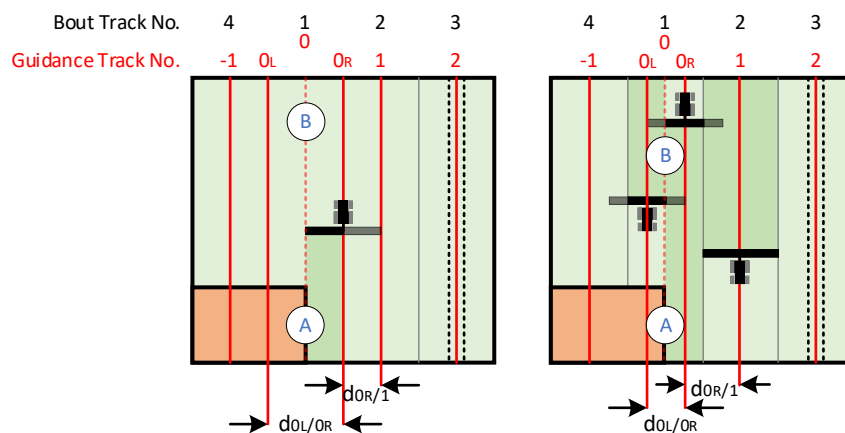


Figure 16: Distance between Track 0_R and 1, and Track 0_R and 0_L

Tramline Control

If a full Working Width shall be used for Track 0 in one working step, the distance $d_{0R/1}$ between Track 0_R and 1 is equal to the Working Width of the implement. In this case the distance $d_{0R/0L}$ between 0_R and 0_L results in the value 0.

In all other cases the resulting Working Width of adding Track 0_R and 0_L shall cover 100% Working Width in order to fulfil equidistant distances between all Bout Tracks.

The distance between Track Number 0_R and 1 depends on the Working Width of the Implement to the time of starting the work. Additionally, this value depends on the implements properties, how a Working Width different to complete Width can be applied. One option may be working with the complete right-hand side or left-hand side. The other option would be to work with the middle of the Implement and turn off the most right and most left side.

The Implement is responsible for the correct calculation of these distances in respect of the Implements properties. The distance $d_{0R/1}$ between Track Number 0_R and 1 and the distance $d_{0R/0L}$ between Track Number 0_R and 0_L are provided by the Implement. If provided in the DDOP, these DDIs shall be listed only once.

To preserve misleading numbering and having always full Working Widths between the Guidance Tracks, this specification requires for tramline level 2, that the Task Controller can handle two Tracks with Number 0, the Tracks 0_R and 0_L . In this document the terms 0_R and 0_L are used, in the communication protocol only one value 0 is used.

2.2.24DDI 669 Track Number Shift

Sometimes the calculated Tramline pattern needs to be shifted. For example, if an obstacle (tree) is in the field or the A-B line is not the starting point for the tramline calculation.

The implement may have the possibility to give a Track Number Shift in its calculation to shift the Tramline pattern. In that case the Tramline calculation is based on the following formula: Track Number = Guidance Track Number + Track Number Shift. The Task Controller shall store track number shift in conjunction with the Unique A-B Guidance Reference Line ID. The TC shall send this value on change of Unique A-B Guidance Reference Line ID to inform the implement that it needs to take this Track Number Shift into account. Therefore, this DDI shall be settable. The Guidance Track Numbers are not influenced by this DDI. A positive Track Number Shift will shift the Tramline pattern in the direction of the lower Guidance Track Numbers. A negative Track Number Shift will shift the Tramline pattern in the direction of the higher Guidance Track Numbers.

Without track number shift, the A-B represents the Track Number 0.

Track Number = **Guidance Track Number** + Track Number Shift.

Tramline Control

The following examples and pictures show the influence of the Track Number Shift. Working Direction is 1 (working from left to right, compared to AB-reference line)
Example Track Number Shift = 0

- Track Number (0) = Guidance Track Number (+0) + (0)
- Track Number (5) = Guidance Track Number (+5) + (0)
- Track Number (-4) = Guidance Track Number (-4) + (0)

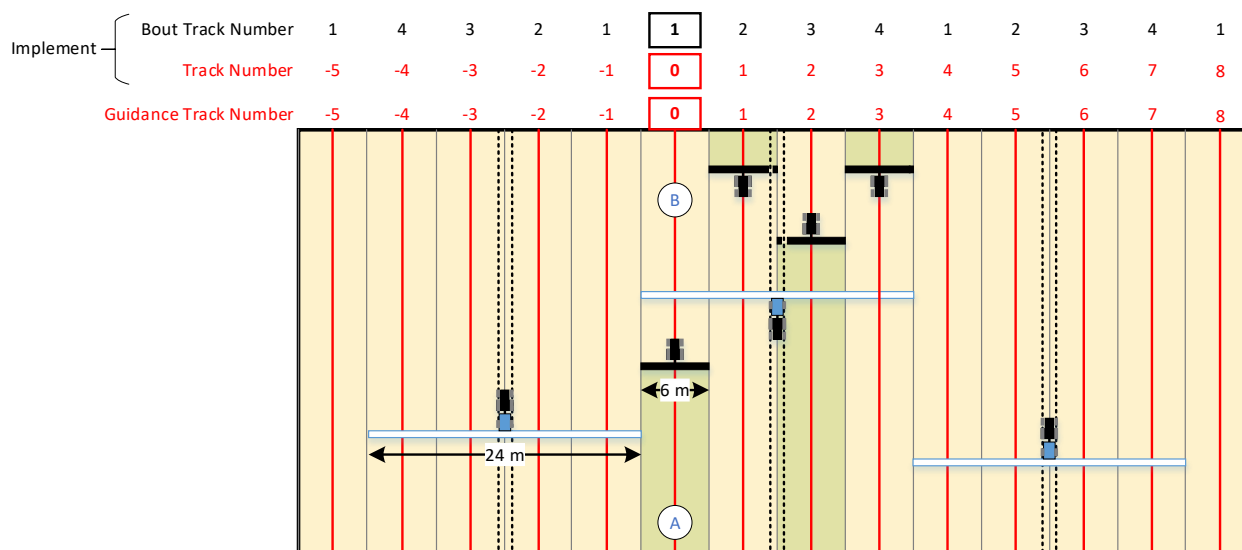


Figure 17: Track Number Shift = 0

Example Track Number Shift = 1

- Track Number (0) = Guidance Track Number (-1) + (1)
- Track Number (5) = Guidance Track Number (+4) + (1)
- Track Number (-4) = Guidance Track Number (-5) + (1)

Tramline Control

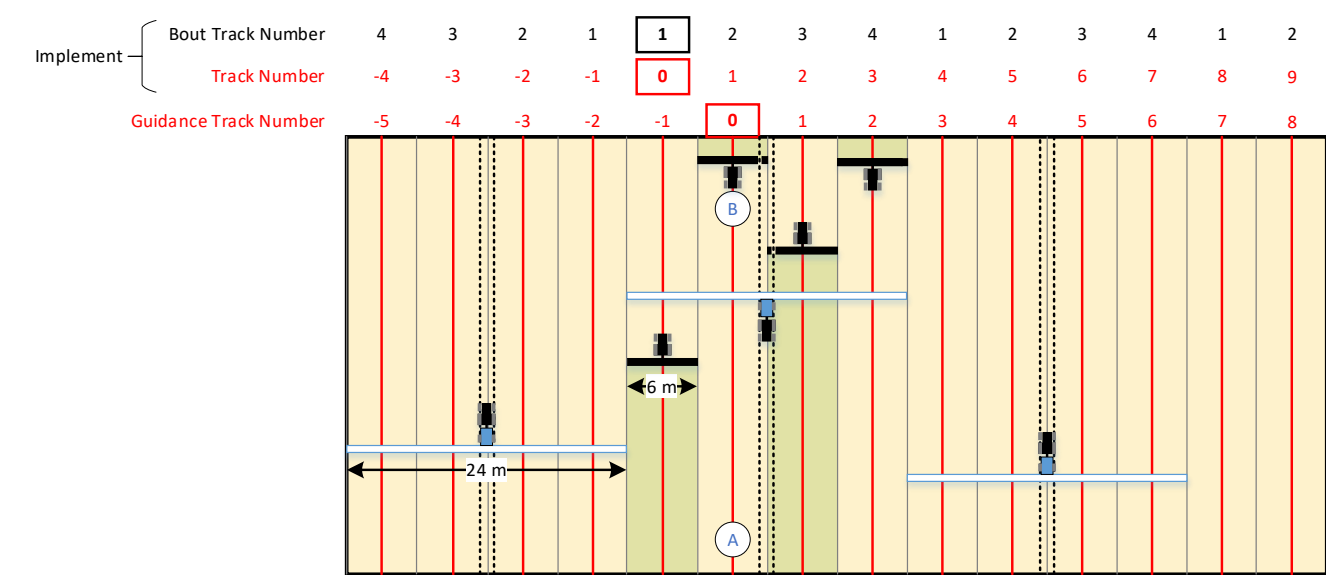


Figure 18: Track Number Shift = 1

Example Track Number Shift = -1

- Track Number (0) = Guidance Track Number (1) + (-1)
- Track Number (5) = Guidance Track Number (+6) + (-1)
- Track Number (-4) = Guidance Track Number (-3) + (-1)

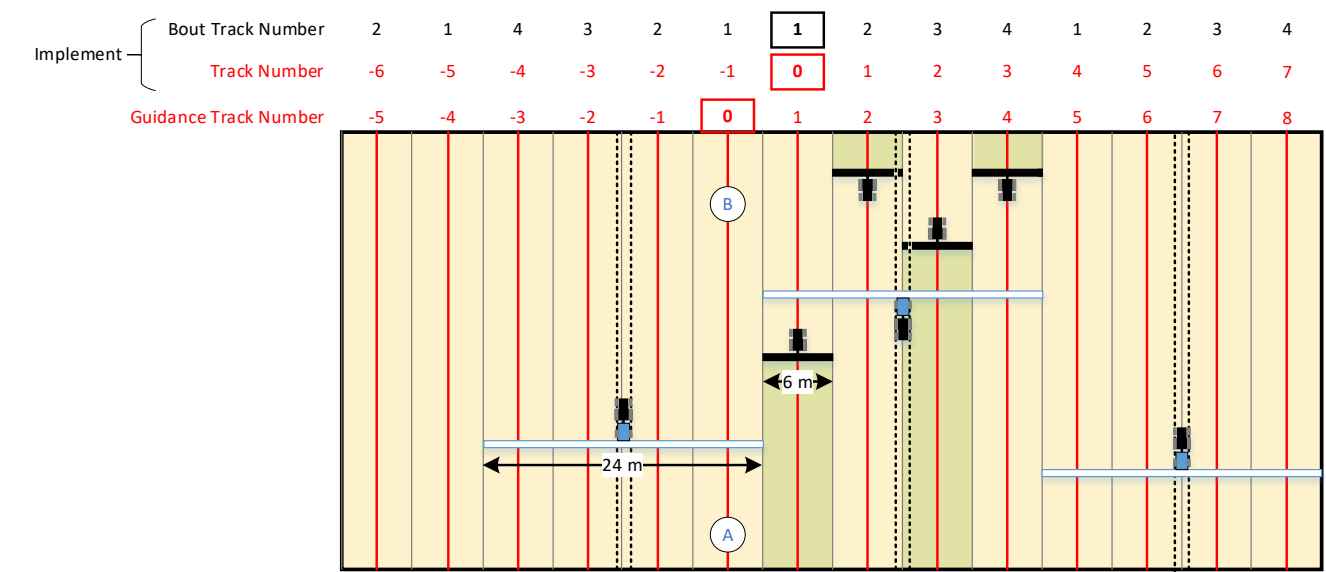


Figure 19: Track Number Shift = -1

Tramline Control

The Task Controller also may offer a possibility to create a shift in its tracks. For example to reuse an already existing A-B line. This Task Controller shift shall be compensated with the Track Number Shift of the implement. For example, the Implement has currently a Track Number Shift of 0, the operator sets in the Task Controller the shift of the tracks to -3, the Task controller shall send a Track Number Shift of -3 to the implement. The numbering of the Guidance Tracks Numbers of the Task Controller shall not be influenced by the Track Number Shift.

The following picture shows an example, where an existing A-B Guidance Reference Line is reused for a second part of the field. In this example, the existing A-B Guidance Reference Line on Guidance Track Number 0 is reused on Track Number 3. Therefore, the TC will set the Track Number Shift to -3.

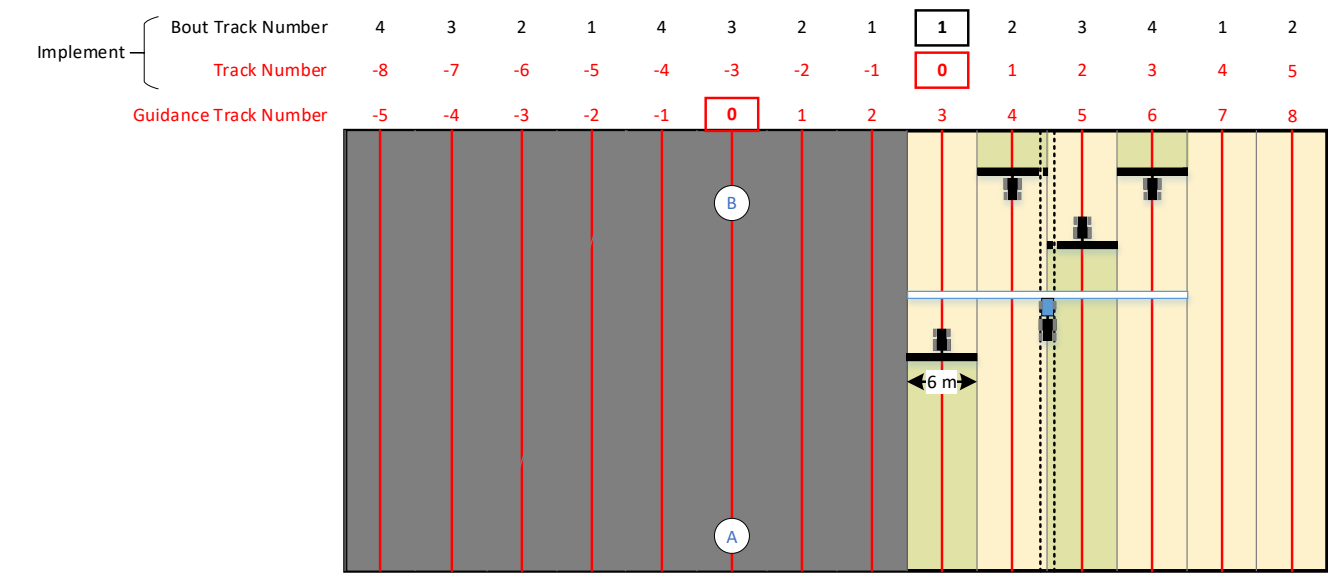


Figure 20: Track Number Shift = 3 This value shall be stored by the Task Controller. See also 2.1.1

Tramline Control

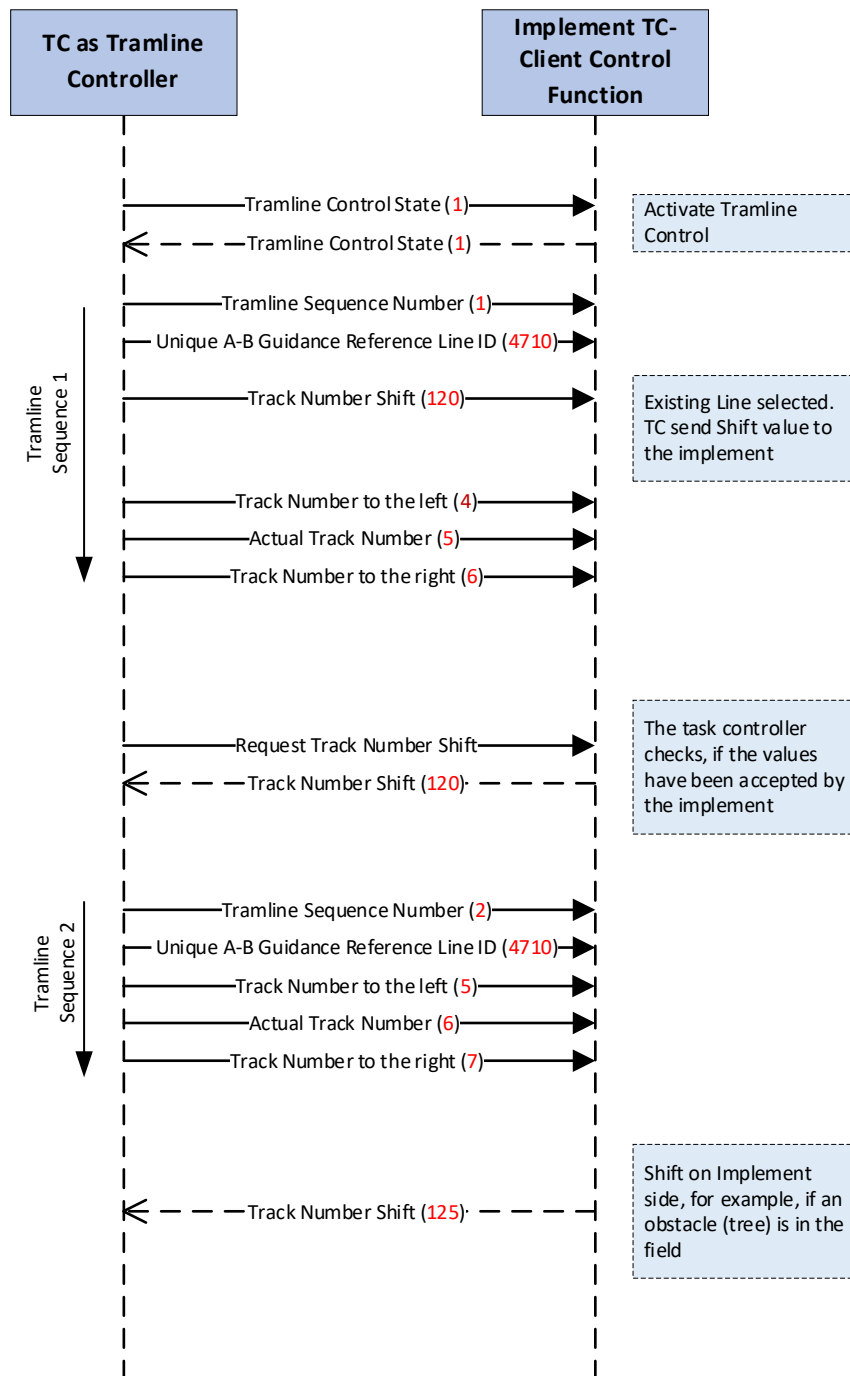


Figure 21: Sequence diagram Track Number Shift

Tramline Control

2.3 Steps to set up a basic Tramline Control System

To have the Tramline System ready to work, the following steps have to be carried out.

2.3.1 Set up Tramline Control Level

The Tramline Control Level has to be set up between Client and Server to define who has to calculate the Tramlines.

2.3.2 Set up Working Widths

The Working Width of the implement and the Working Width of the Sprayer/Fertilizer have to be defined to set the distances between the Guidance Tracks and Tramline Tracks.

2.3.3 Set up Track Number 0 (A-B line) position

Next to the Working Widths there has to be an A-B line. This line may be identified by the Unique A-B Guidance Reference Line ID.

This A-B line is defined as Track Number 0 by default.

The task controller may set any other guidance track as Track Number 0. If he do so, the Task Controller informs the implement via the DDI Track Number Shift over the resulting movement. This Track Number 0 is the starting point of the work for the rest of the field.

2.3.4 Set up Working Width for the first Bout

The Working Width the user wants to start with, together with the property of the Implement to fulfil the wanted working width, results in the distance $d_{0R/1}$ between Track 0_R and 1. This distance results therefore in the distance $d_{0R/0L}$ between the Tracks 0_R and 0_L . See also section 2.2.23.

3 ANNEX - Examples for Tramline Control

In the following sections are detailed out some examples for different use cases for Tramline Control.

3.1 Asymmetric Tramlines, starting with 100% Working Width

The Seeder with 6m Working Width starts working with a full Working Width (100%, 6m) on Guidance Track Number 0 in direction of A-B line. The Working Direction is 1, which means the Field Border is on the left-hand side of the Implement and the Guidance Track Numbering is increasing from left to right in A-B direction. The Sprayer has a Working Width of 24m. This results in a Tramline Rhythm of 4 Bout Tracks and an asymmetric Tramline on Bout Track Numbers 2 and 3. The Seeder will create a half Tramline on each Bout Number 2 and 3.

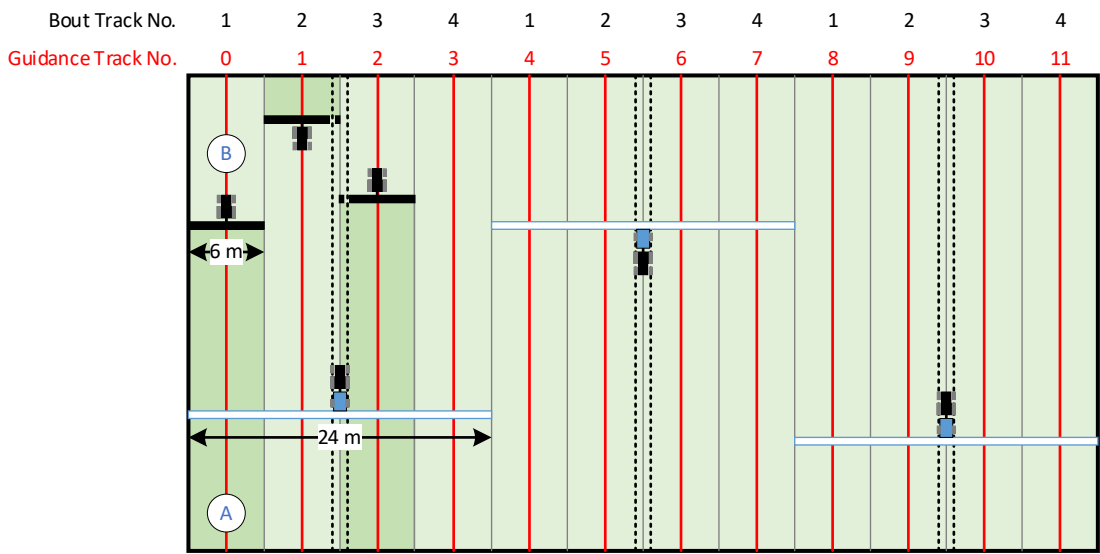


Figure 22: Asymmetric Tramlines, Seeder starts with 100% Working Width

The following picture shows a possible sequence diagram for the needed parameters.

Tramline Control

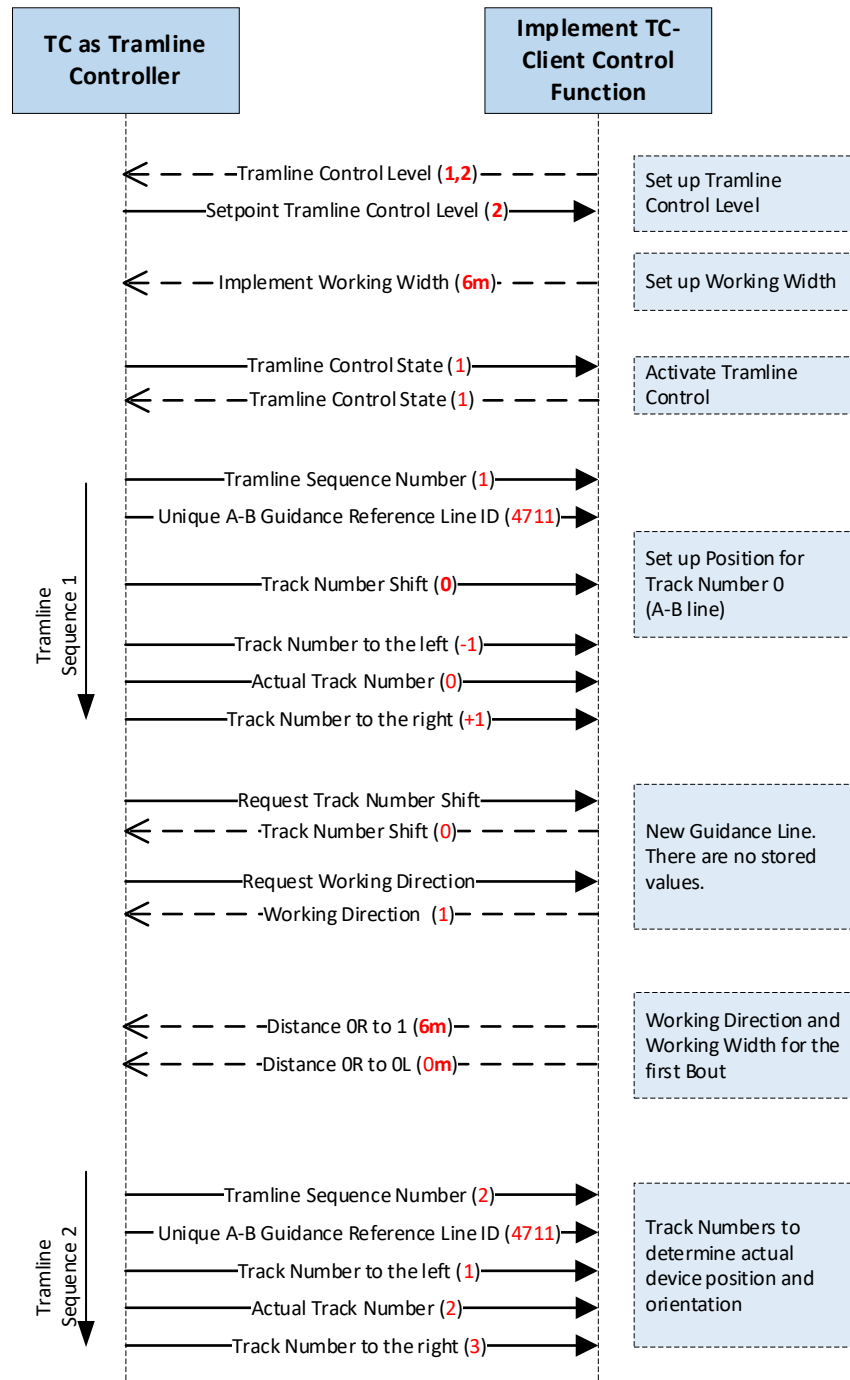


Figure 23: Sequence diagram - Asymmetric Tramlines, Seeder starts with 100% Working Width

3.2 Symmetric Tramlines, starting with 50% Working Width (left side working)

The Seeder with 6m Working Width starts working with a half Working Width (50%, 3m) on Guidance Track Number 0_R in direction of A-B line. The Working Direction is 1, which means the Field Border is on the left-hand side of the Implement and the Guidance Track Numbering is increasing from left to right in A-B direction. The Sprayer has a Working Width of 24m. This results in a Tramline Rhythm of 4 Bout Tracks and a symmetric Tramline on Bout Track Number 3. The Seeder will create a Tramline on Bout Number 3.

Tramline Control

The Guidance Track Number 0 has to be filled up to a complete Seeder Working Width of 6m in two separate working steps. For that, there have to be two additional Tracks (0_R and 0_L) for the Track Number 0.

In this case, the Seeder works both Tracks (0_R and 0_L) with half Working Width on the complete left side of the Implement. The distance $d_{0R/1}$ between Track Number 0_R and 1 is 3m. On the other side of Track Number 0, the Task Controller has to create the line number 0_L . The distance $d_{0R/0L}$ of Track 0_R and 0_L is 6m.

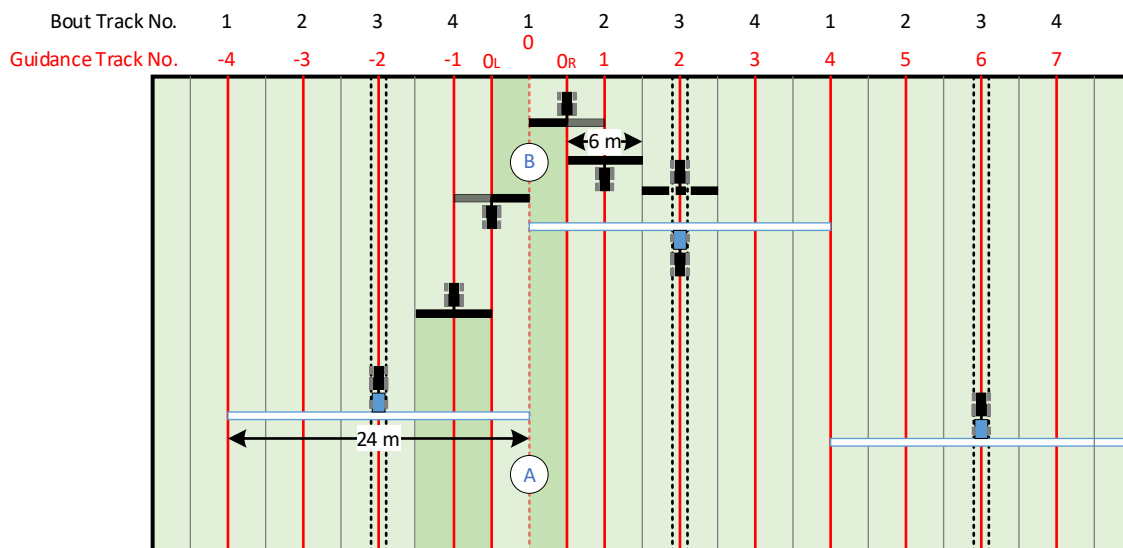


Figure 24: Symmetric Tramlines, Seeder starts with 50% Working Width (left or right-side working)

The following picture shows a possible sequence diagram for the needed parameters.

Tramline Control

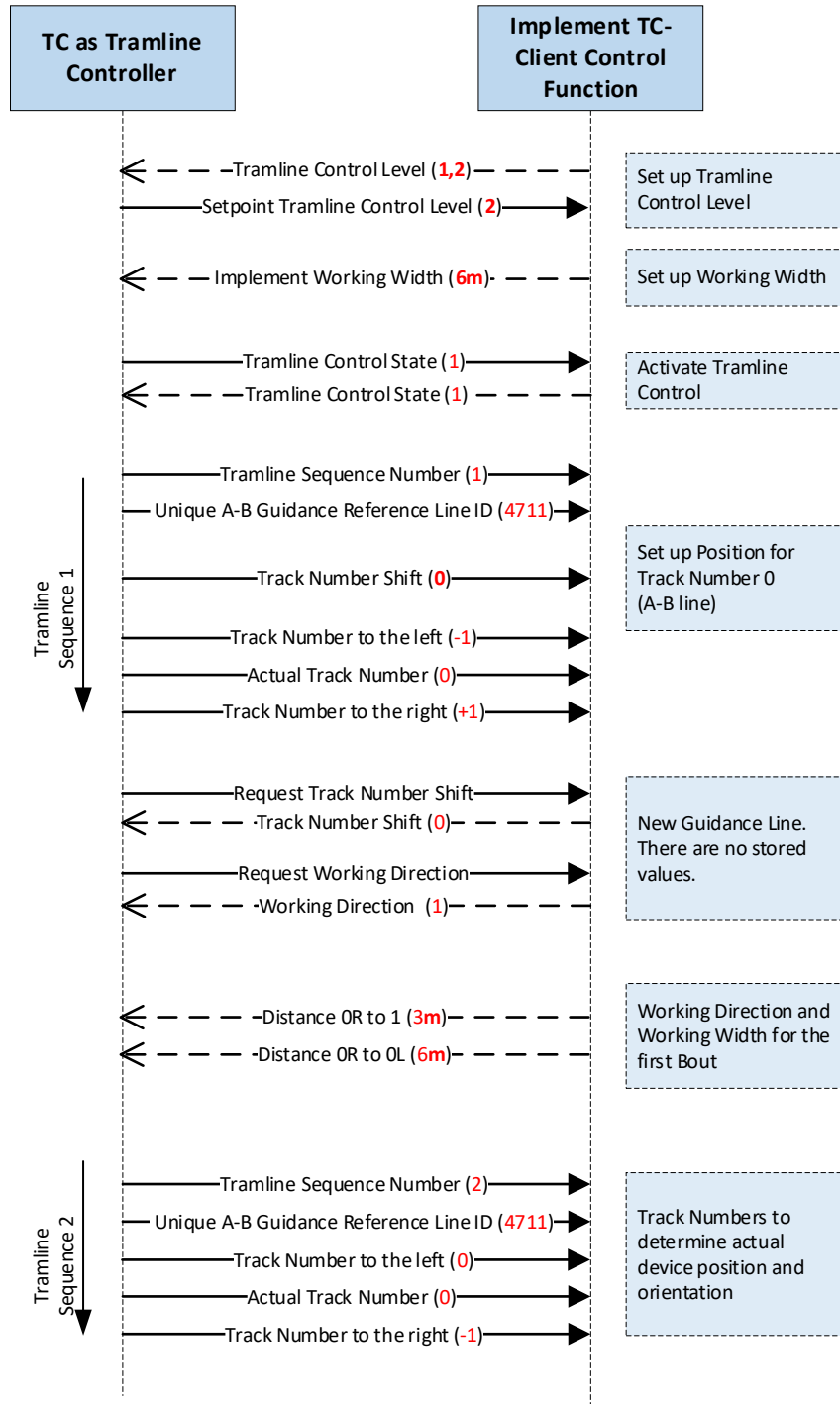


Figure 25: Sequence diagram - Symmetric Tramlines, Seeder starts with 50% Working Width (left or right-side working)

3.3 Symmetric Tramlines, starting with 50% Working Width (middle working)

This is the same example as the described before. The difference is the property of the Implement to create a half Working Width.

In this case, the Seeder works both Tracks (0_R and 0_L) with half Working Width on the middle of the Implement. For that, the distance $d_{0R/1}$ between Track Number 0_R and 1 is 4.5m. On the other side

Tramline Control

of Track Number 0, the Task Controller has to create the line number 0_L . The distance $d_{0R/0L}$ between Track 0_R and 0_L is 3m.

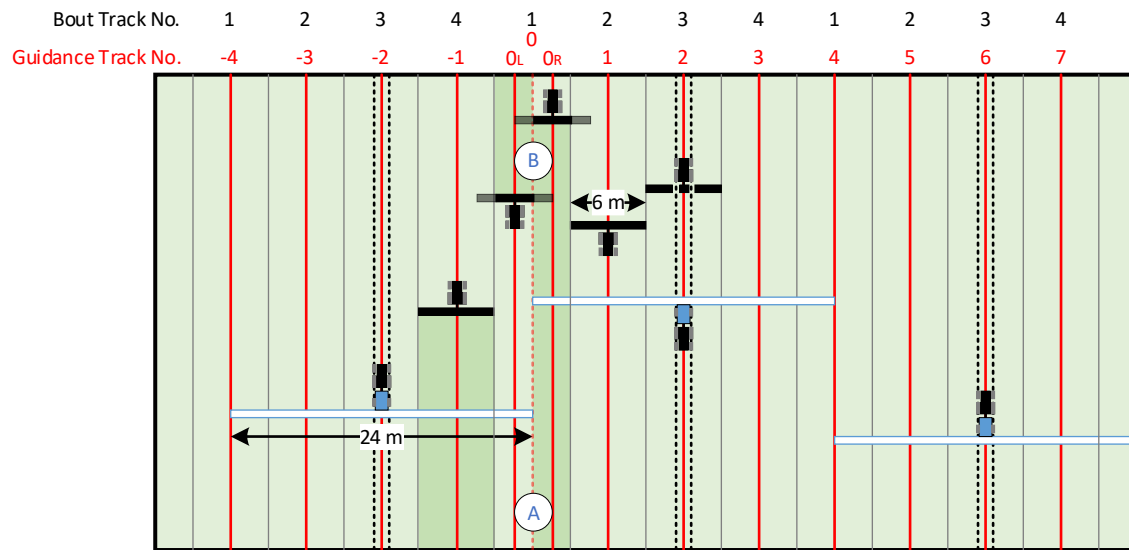


Figure 26: Symmetric Tramlines, Seeder starts with 50% Working Width (middle working)

The following picture shows a possible sequence diagram for the needed parameters.

Tramline Control

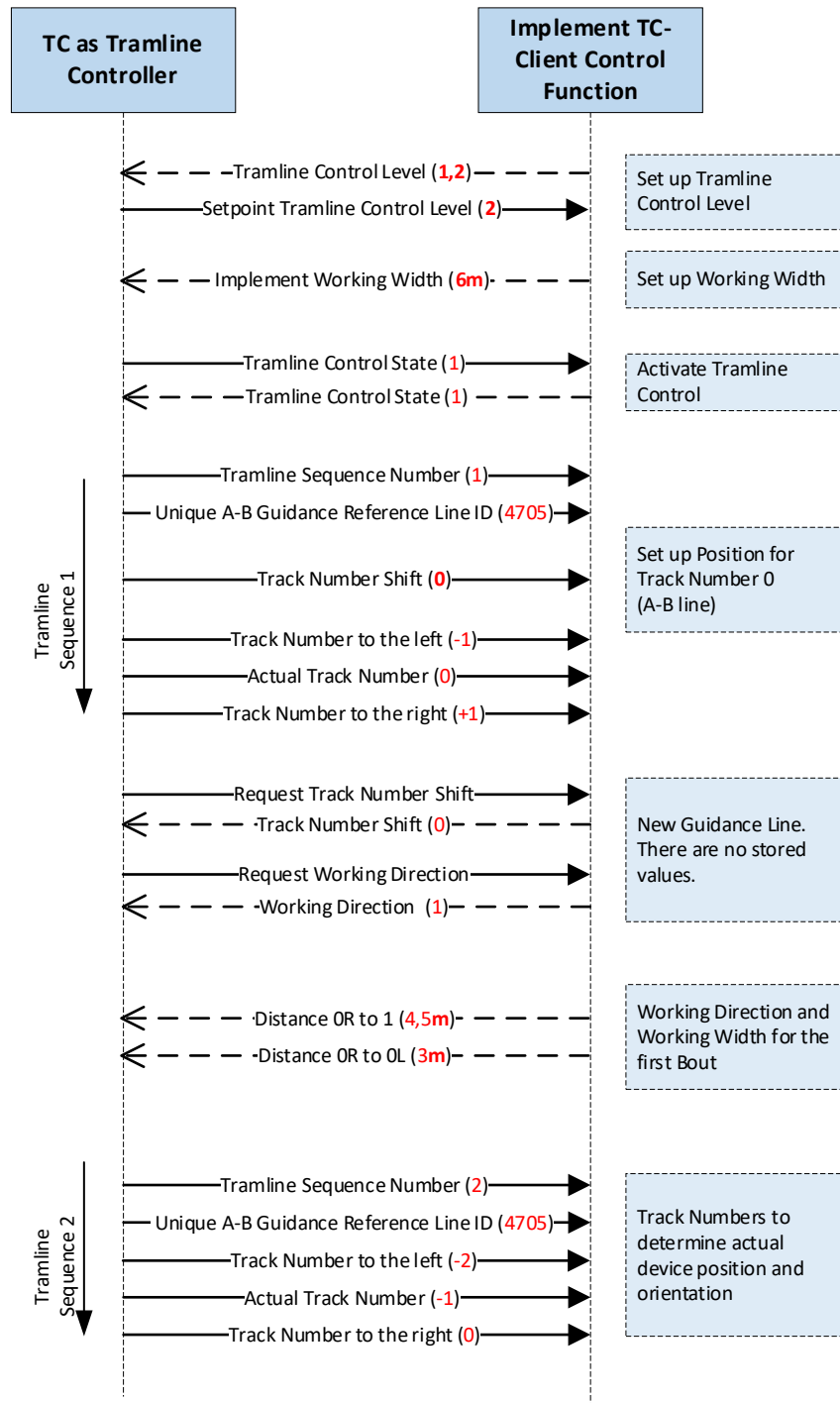


Figure 27: Sequence diagram - Symmetric Tramlines, Seeder starts with 50% Working Width (middle working)

3.4 Asymmetric Tramlines, first finish surrounding headland, then the main field

This example shows the situation where first 3 complete rounds in a surrounding headland are sown. For each round at first the area in Pattern A, then the Pattern B, then Pattern C and then Pattern D. This is done 3 times to fill up the surrounding headland area. Then the main field is finished, based on Pattern A. Every part of the field (Pattern) has its own A-B line.

Tramline Control

For this example, it is required to communicate the correct A-B Guidance Reference Line ID to the Implement.

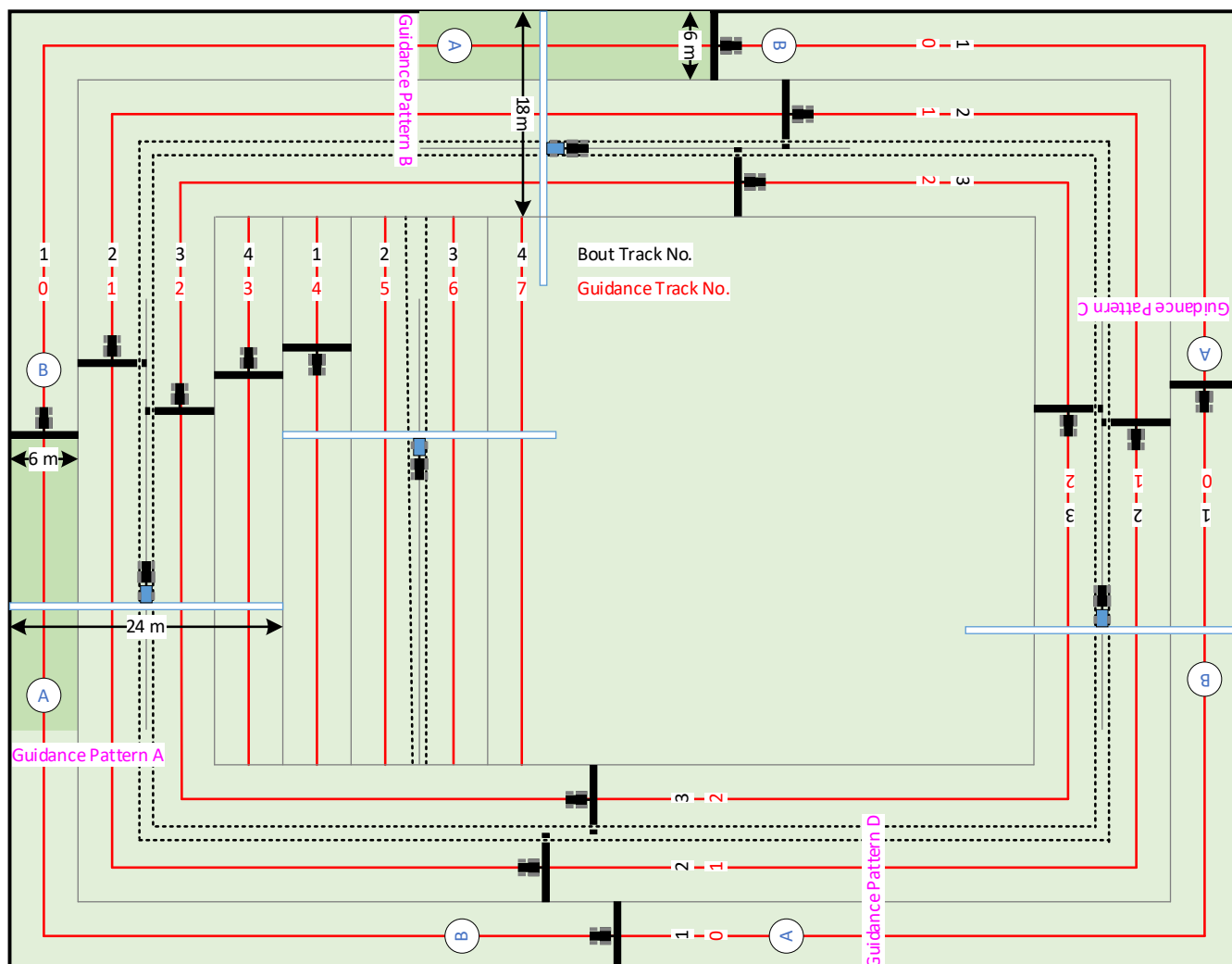


Figure 28: Asymmetric Tramlines, first finish surrounding headland, then the main field

The following picture shows a possible sequence diagram for the needed parameters.

Tramline Control

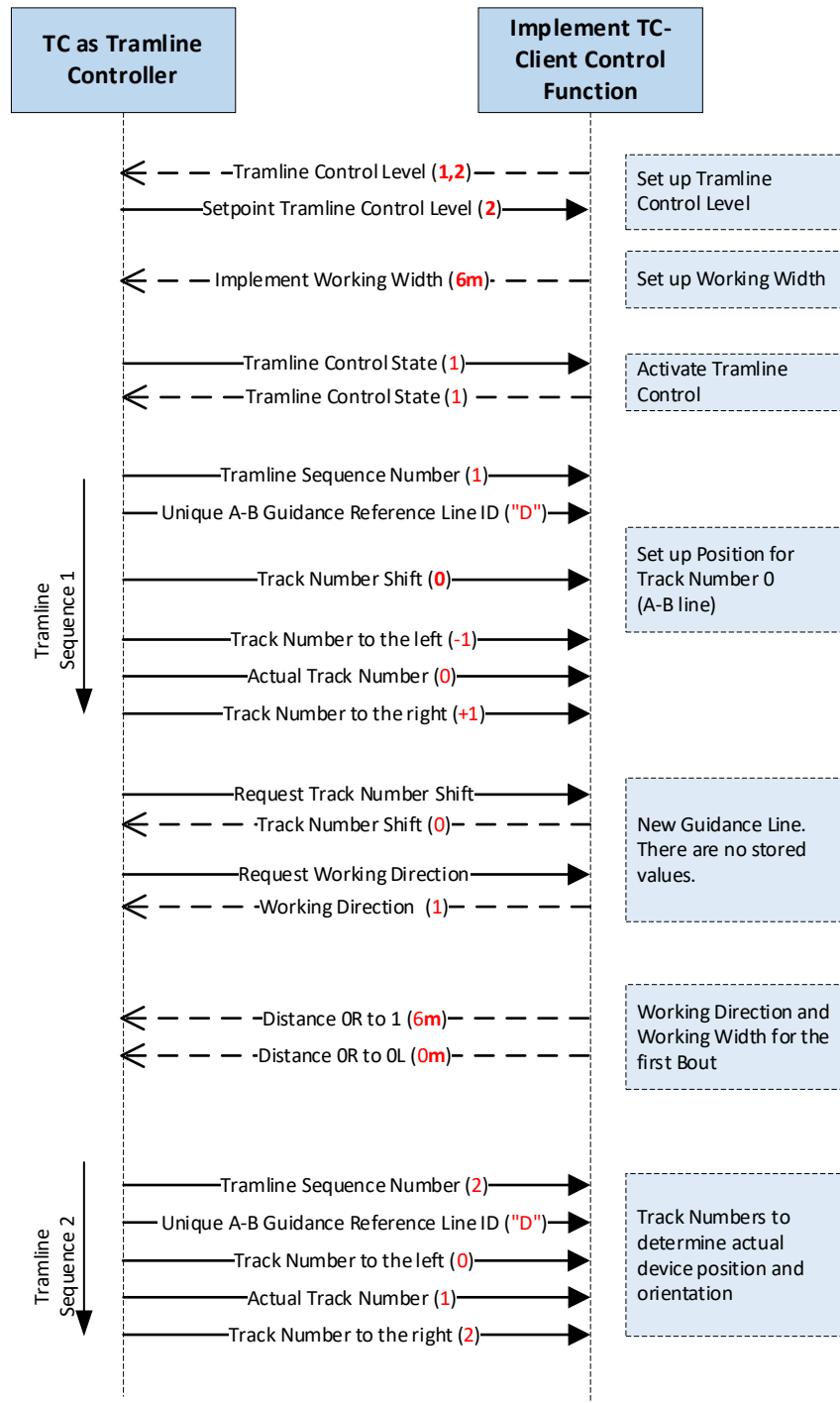


Figure 29: Sequence diagram - Asymmetric Tramlines, first finish surrounding headland, then the main field

Tramline Control

3.5 Tramline Control Level 1 DDOP Examples

3.5.1 Example 1a: Tramline Control DDIs are placed in the device element

Two Section Control Booms with 12 sections and actual tramline states

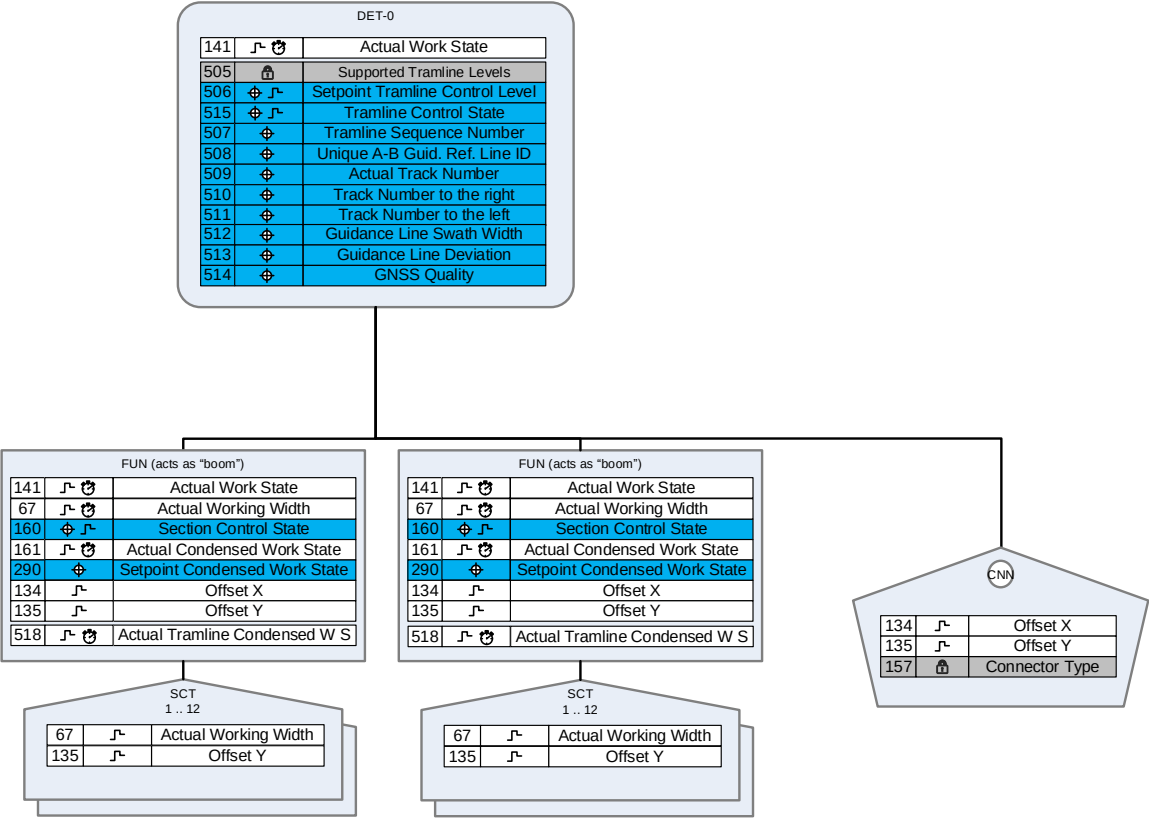


Figure 30: TLC Level 1 DDOP example 1a

Tramline Control

3.5.2 Example 1b: Tramline Control DDIs are placed in a separate function element

Two Section Control Booms with 12 sections and actual tramline states

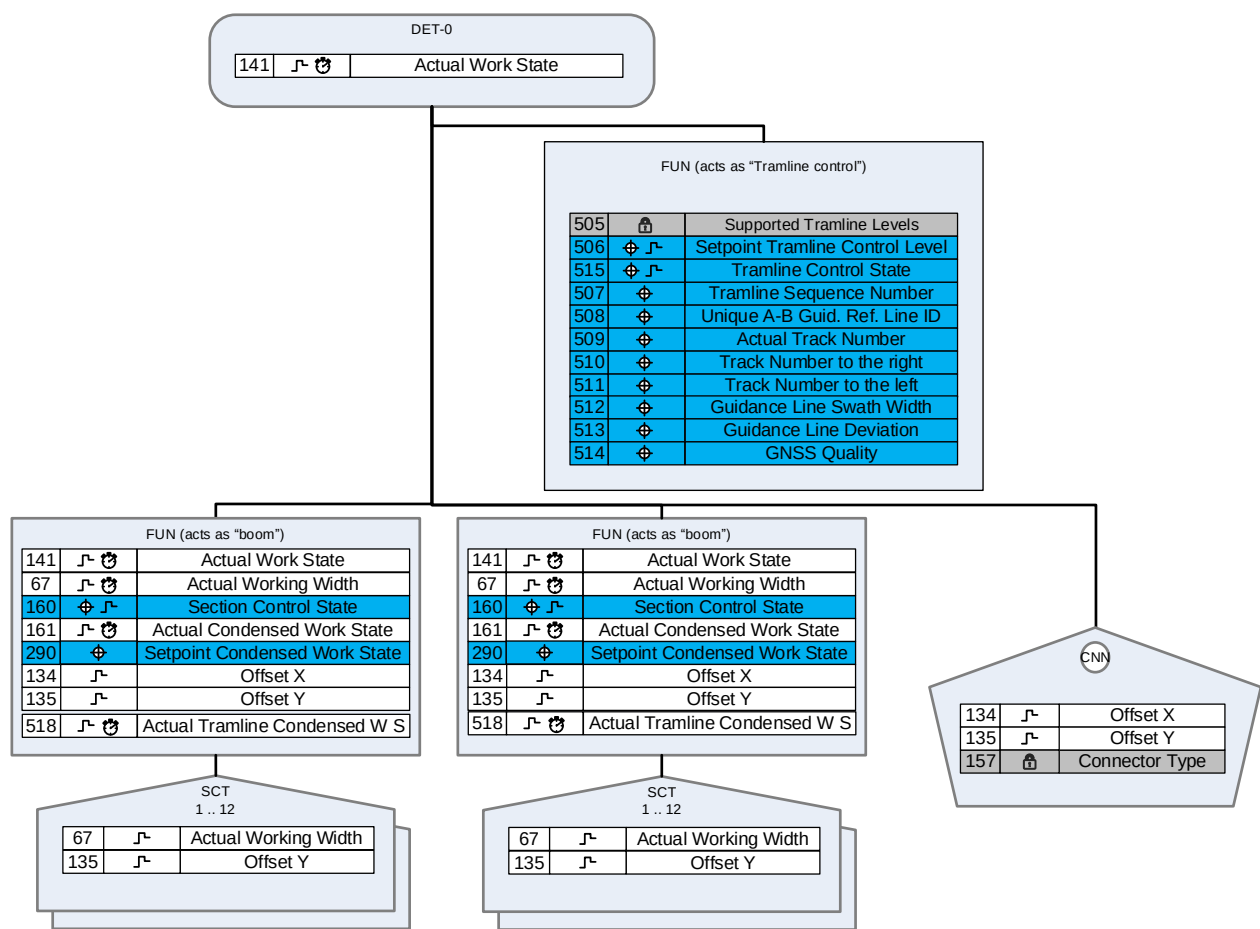


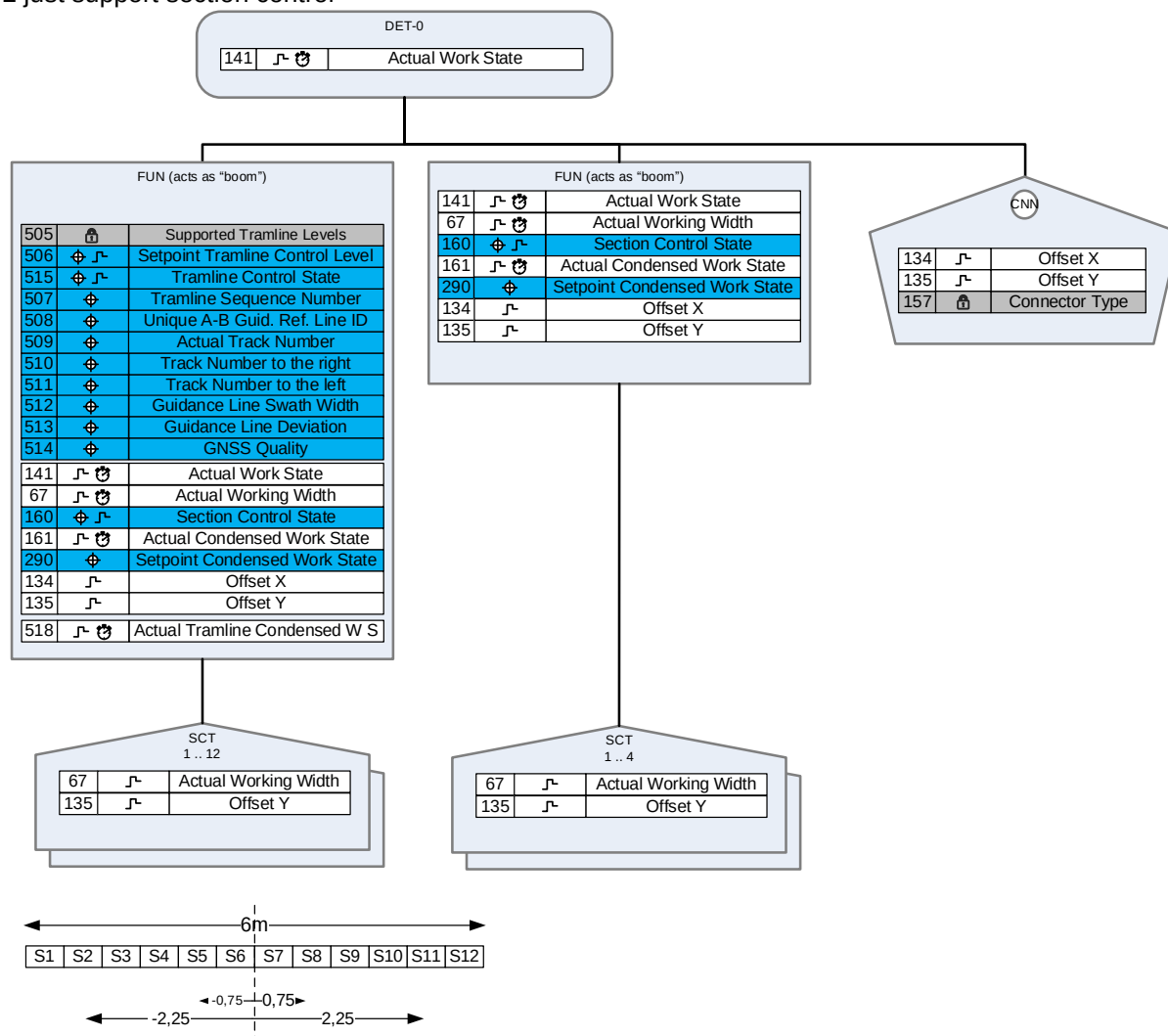
Figure 31: TLC Level 1 DDOP example 1b

Tramline Control

3.5.3 Example 1c: Tramline Control DDIs are placed in a boom element

Boom 1 support section control and actual tramline states

Boom 2 just support section control



Example of a 6m drill, with 12 seeding Sections And 4 fertilizing sections.
The 12 seeding sections can also be used for tramlining.

The tramline wheel track distance is set to 1,5 m.

Sprayer width of 18m:

Rhythm 1 to 3.

Bout 2: "Tramline Sections" S 5 and S 8 are activated.

Sprayer width of 24m:

Working left to right:

Rhythm 1 to 4

Bout 2: "Tramline Sections" S 2 is activated.

Bout 3: "Tramline Sections" S 2 is activated.

Sprayer width of 24m:

Working right to left:

Rhythm 1 to 4.

Bout 2: "tramline section" S 11 is activated.

Bout 3: "tramline section" S 11 is activated.

Sprayer width of 24m Symmetric.

First bout with 6 section activated.

All other bouts all sections are activated.

Bout 3: "Tramline sections" S5 and S8 are activated.

Note:

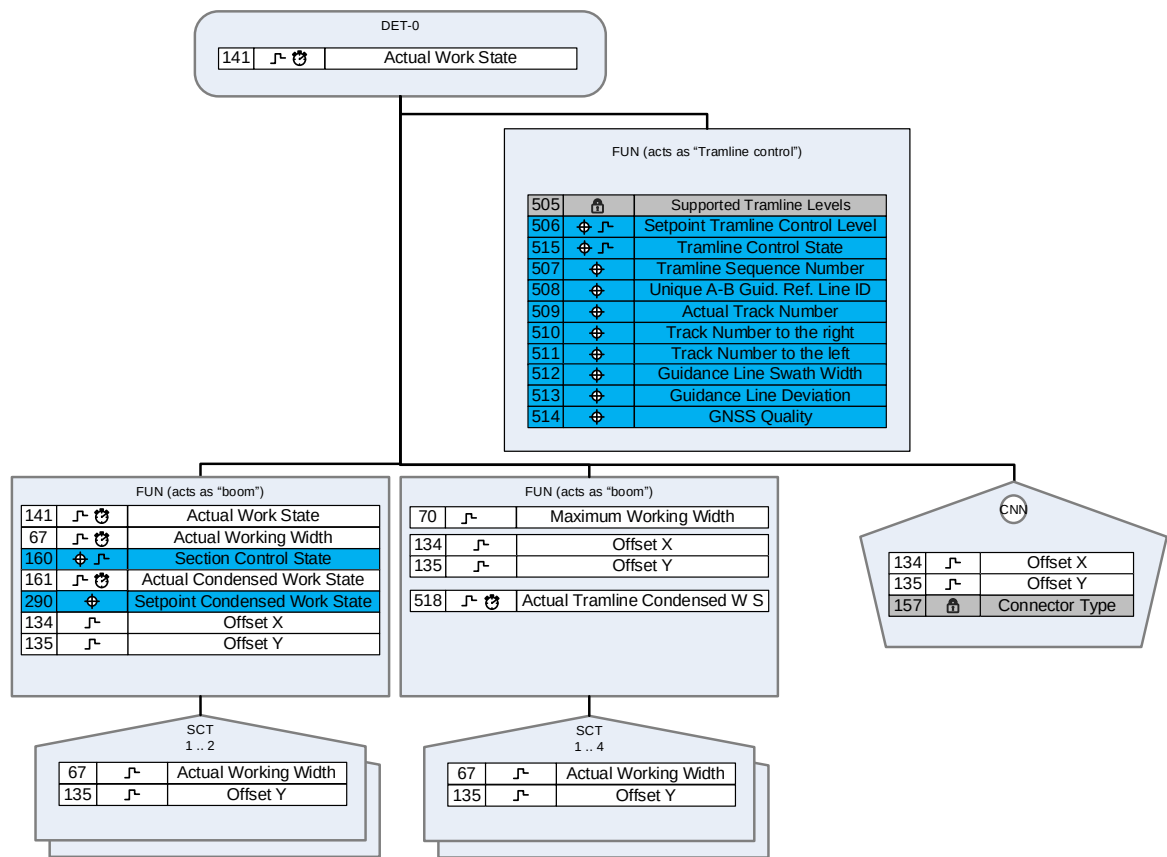
With this configuration many tramline rhythms are possible, because each section can be used as a tramline section.

Figure 32: TLC Level 1 DDOP example 1c

Tramline Control

3.5.4 Example 2: Tramline Control Boom

Boom 1 support section control
Boom 2 reports the actual tramline states



Example of a 6m drill, with 4 tramline sections and 2 sections.

The tramline wheel track distance is fixed to 1,5 m.

Sprayer width of 18m:
Rhythm 1 to 3.
Bout 2 : Tramline Sections TS 2 and TS 3 are activated.

Sprayer width of 24m:
Working left to right:
Rhythm 1 to 4
Bout 2: Tramline Section TS 1 is activated.
Bout 3: Tramline Section TS 1 is activated.

Sprayer width of 24m
Working right to left:
Rhythm 1 to 4.
Bout 2: Tramline Section TS 4 is activated.
Bout 3: Tramline Section TS 4 is activated.

Sprayer width of 24m Symmetric:
First bout with one section activated.
All other bouts both sections are activated.
Bout 3: Tramline Section TS2 and TS3 are activated.

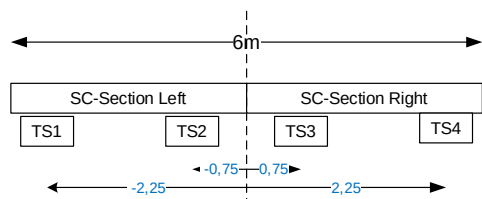


Figure 33: TLC Level 1 DDOP example 2

Tramline Control

3.6 Examples for Tramline Control Level 3

This chapter describes a solution where the Tramline Controller is responsible for calculating the Tramline Tracks. The Tramline Controller will command the Implement to turn Tramline sections on or off.

A level 3 Tramline Controller will be able to command the following implements to apply tramlining:

- Implements without section control sections, but with a boom with one or more tramline sections. (Pneumatic seeders)
- Implements with a boom with section control sections and a second boom for tramline sections. (Pneumatic seeders)
- Implement with one boom where the section control sections can be used for tramlining. (Precision seeders)

A mandatory requirement for the implement is, is that it shall be able to switch each tramline section independently.

A Tramline controller needs the following information from an implement to apply proper tramlines in the field:

- Location of the tramline sections on the implement
- The working width of the tramline section

For Level 3 tramlining the following DDI's will be used:

- Tramline Control State
- Setpoint Tramline Condensed Section State
- Actual Tramline Condensed Section State

Work flow:

At start up the Implement uploads its DDOP to the Task Controller. If the Task Controller detects a DDI with the Tramline Level set to 3, it knows the implement supports Tramline commands.

The Tramline Controller provides a screen where the operator can enter the following information:

- Sprayer width
- Tramline Wheel Track Distance
- etc.

When the operator has entered this information, the Tramline Controller verifies if this information match with the information provided in the DDOP of the Implement. It is the Tramline Controller's responsibility to inform the operator in case of any incompatibility.

Example:

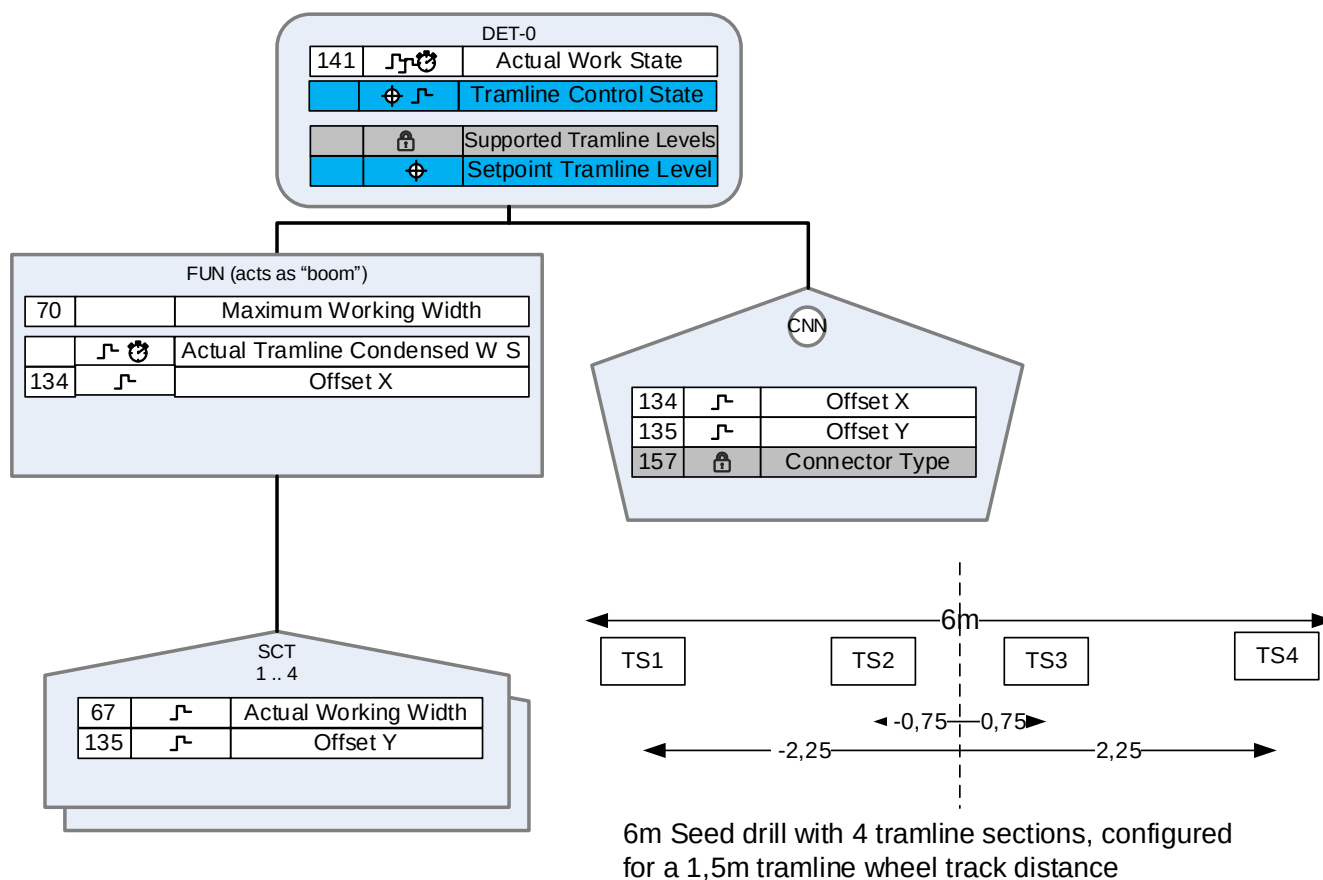
The Tramline Wheel Distance is set to 1,5m but the Tramline valves on the implement are configured for a 2 m Tramline Wheel Distance.

When the farmer has entered a valid configuration, the Tramline Controller will be allowed to send Tramline Condensed Section States to the implement.

The following pictures show different examples of DDOPs for Tramline Control.

Tramline Control

Case 1: TC-TRC DDOP Example. DDOP without sections control sections but with Tramline Sections.



Example of a 6m drill, with 4 tramline sections.

The tramline wheel track distance is fixed to 1,5 m.

Sprayer width of 18m:

Rhythm 1 to 3.

Bout 2: Tramline Sections TS 2 and TS 3 are activated.

Sprayer width of 24m:

Working left to right:

Rhythm 1 to 4.

Bout 2: Tramline Section TS 1 is activated.

Bout 3: Tramline Section TS 1 is activated.

Sprayer width of 24m:

Working right to left:

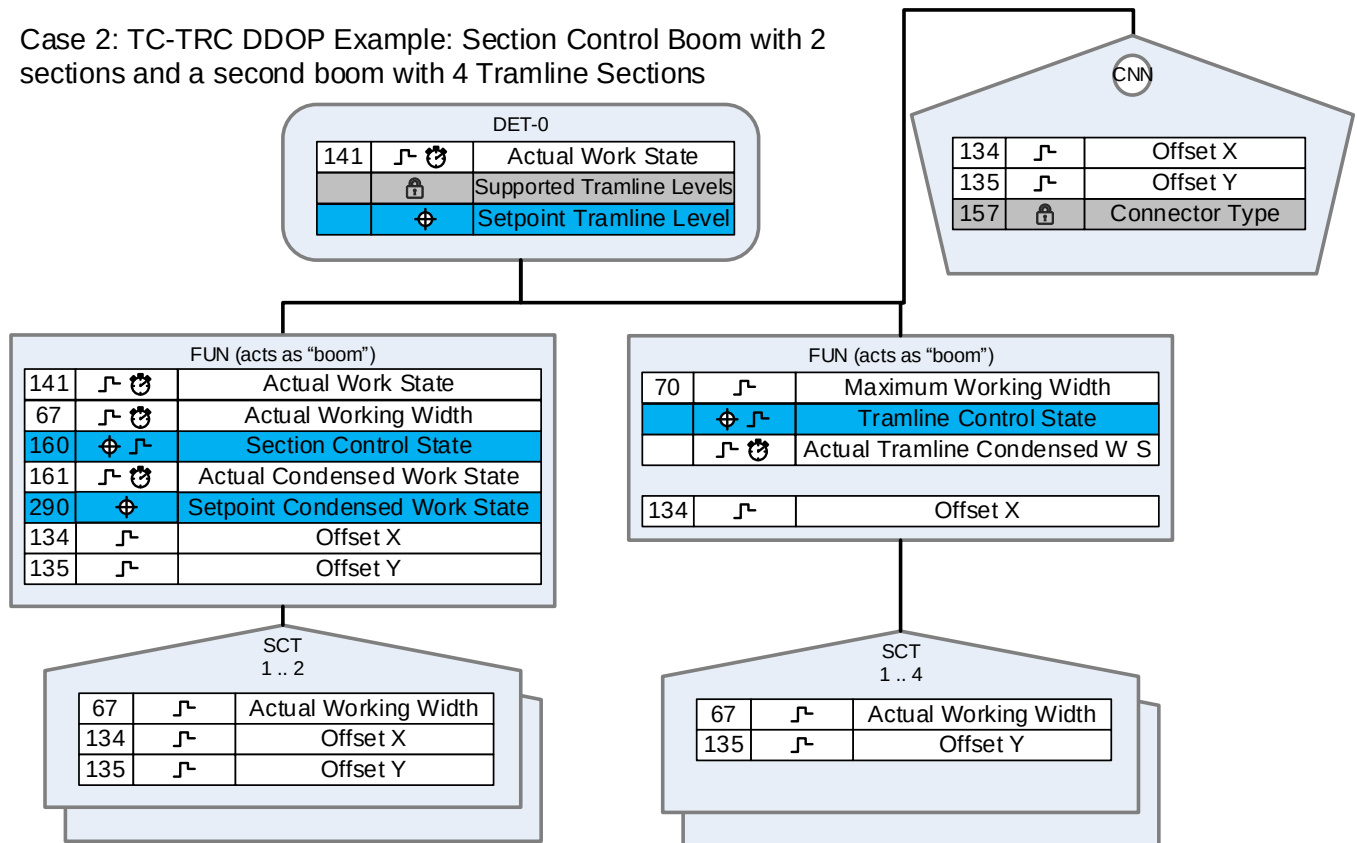
Rhythm 1 to 4

Bout 2: Tramline Section TS 4 is activated.

Bout 3: Tramline Section TS 4 is activated.

Tramline Control

Case 2: TC-TRC DDOP Example: Section Control Boom with 2 sections and a second boom with 4 Tramline Sections



Example of a 6m drill, with 4 tramline sections and 2 sections.

The tramline wheel track distance is fixed to 1,5 m.

Sprayer width of 18m:

Rhythm 1 to 3.

Bout 2 : Tramline Sections TS 2 and TS 3 are activated.

Sprayer width of 24m:

Working left to right:

Rhythm 1 to 4

Bout 2: Tramline Section TS 1 is activated.

Bout 3: Tramline Section TS 1 is activated.

Sprayer width of 24m

Working right to left:

Rhythm 1 to 4.

Bout 2: Tramline Section TS 4 is activated.

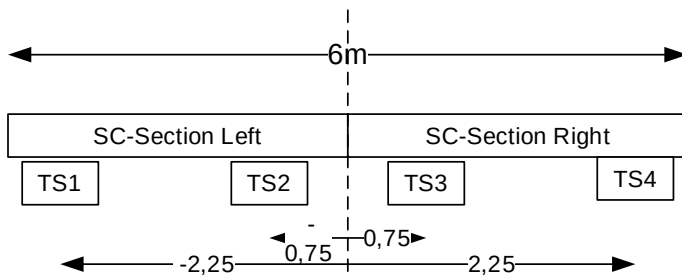
Bout 3: Tramline Section TS 4 is activated.

Sprayer width of 24m Symmetric:

First bout with one section activated.

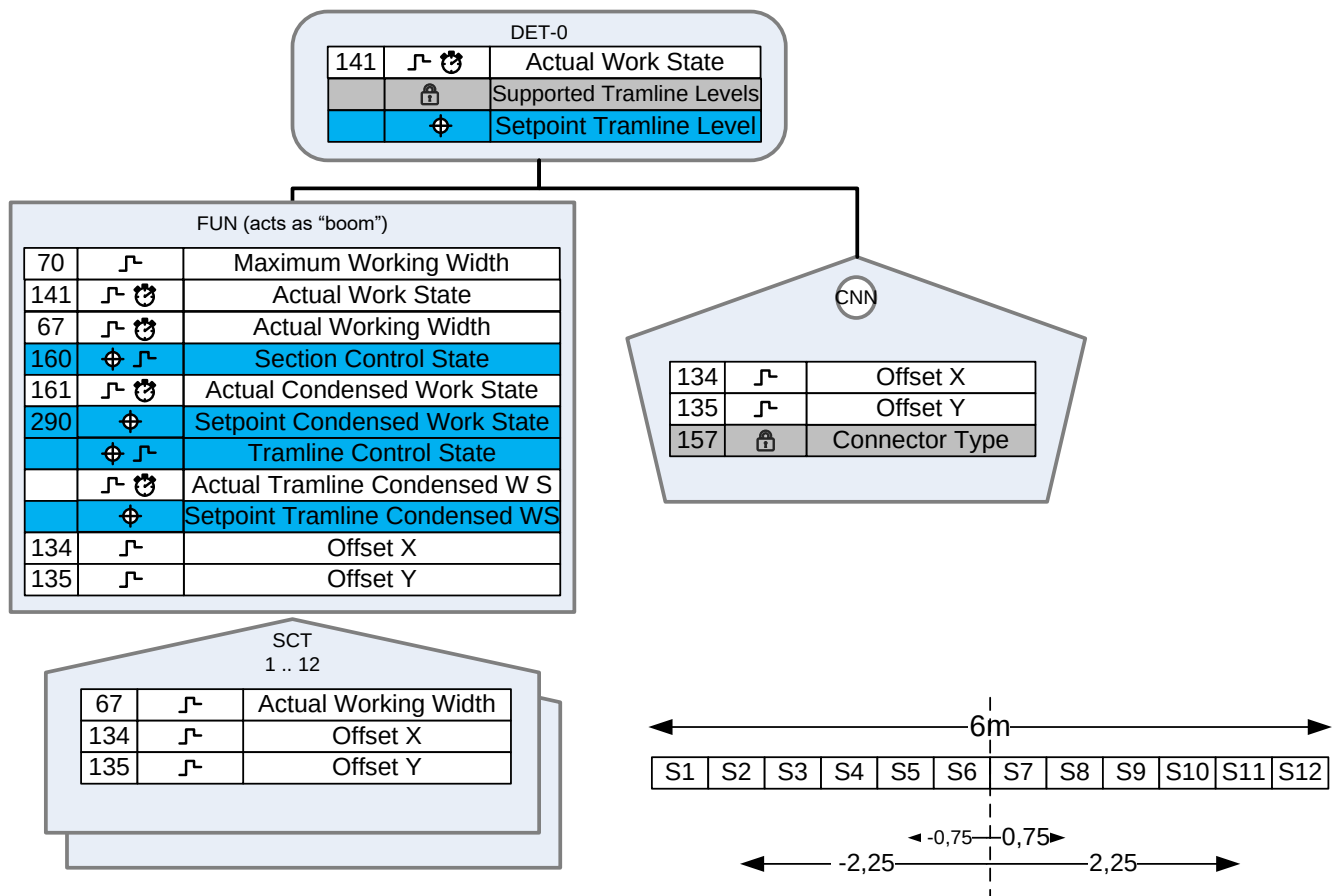
All other bouts both sections are activated.

Bout 3: Tramline Section TS2 and TS3 are activated.



Tramline Control

Case 3: TC-TRC DDOP Example. Section Control Boom with Tramline Sections



Example of a 6m drill, with 12 sections, which can also be used for tramlining.

The tramline wheel track distance is set to 1,5 m.

Sprayer width of 18m:

Rhythm 1 to 3.

Bout 2: "Tramline Sections" S 5 and S 8 are activated.

Sprayer width of 24m:

Working left to right:

Rhythm 1 to 4

Bout 2: "Tramline Sections" S 2 is activated.

Bout 3: "Tramline Sections" S 2 is activated.

Sprayer width of 24m:

Working right to left:

Rhythm 1 to 4.

Bout 2: "tramline section" S 11 is activated.

Bout 3: "tramline section" S 11 is activated.

Sprayer width of 24m Symmetric.

First bout with 6 section activated.

All other bouts all sections are activated.

Bout 3: "Tramline sections" S5 and S8 are activated.

Note:

With this configuration many tramline rhythms are possible, because each section can be used as a tramline section.